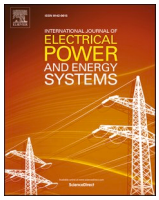
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Linearized power flow calculation of bipolar DC distribution network with multiple flexible equipment

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ABSTRACT

The bipolar direct current distribution network (BDCDN) has the advantages of high reliability, large transmission, and high efficiency. However, voltage unbalance is a crucial power quality problem of BDCDN. On one hand, flexible equipment, such as DC electric spring, DC power flow controller, and DC transformer, can effectively suppress unbalanced voltage. On the other hand, the diversity of topology and control methods of multiple types of flexible equipment complicates power flow calculation. In this study, a linear power flow algorithm is proposed. First, the structural characteristics of a BDCDN with multiple types of flexible equipment are analyzed. Second, a linear ZIP load model based on Taylor expansion is derived. The steady-state model of multiple types of flexible equipment is established by analyzing the topological structure and control mode. Third, a network modeling and power flow calculation method of BDCDN with multiple types of flexible equipment are proposed. Finally, the modified IEEE14-node case and IEEE33-node case are used to verify the validity and correctness of the proposed algorithm. The effectiveness of flexible equipment on the power quality issue is demonstrated through comparison studies of power flow distributions.

1. Introduction

Given the increase of distributed generation capacity and DC loads, such as electric vehicle charging piles, communication equipment, smart data, and computing centers [1], DC power supply technology has been an important development direction in recent years. Compared with the AC distribution network, the DC distribution network has the advantages of higher efficiency, larger system capacity, longer transmission distance, and better power quality [2,3]. The topological structure of the DC distribution network can be divided into unipolar and bipolar networks [4]. The bipolar DC distribution network (BDCDN) has two reliable and flexible voltage levels with large transmission capacity and high overall efficiency [5]. However, the voltage unbalance among different poles is a crucial power quality problem that increases line loss and offsets the positive and negative voltages, thereby affecting the normal load operation [6].

Interconnection equipment, such as DC power flow controllers (DCPFCs) and DC transformers (DCTs), can be connected to the grid side of a BDCDN to suppress voltage unbalance. Compensation equipment, such as DC electric springs (DCESs), can be connected to the load side. In [6], the effectiveness of DCESs in relieving voltage unbalance in BDCDN was verified. In [7], a method for suppressing the unbalanced current of a transmission line based on DCPFCs was proposed, which proved the flexible control effect of DCPFCs on the power flow. In [8], a DCT topology of single and multiple outputs was proposed, which achieved optimal balance control of the positive and negative output port voltages using the adaptive power adjustment control method.

Power flow calculation is the basis of power grid construction planning, operation control, power quality monitoring, and protection configuration [9]. At present, most of the power flow calculation methods for DC distribution networks use iterative algorithms. In [10], it achieved improvements based on the Gaussian algorithm and formed the successive approximation power flow calculation. In [11], based on Newton–Raphson method in the constant power operation mode of voltage source converter, an AC/DC hybrid system power flow algorithm was established. In [12], it analyzed the structure characteristics of the true bipolar high-voltage DC grid and proposed the iterative power flow calculation of the true bipolar high-voltage DC grid considering the different control methods of voltage source converter.

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Nomenclature

Variables

I_{Ppi} Load current of constant power load of node i
V_{pnuj}, V_{nunj}, V_{pnj} Positive, negative, and bipolar voltages of node j
V_{pnuk}, V_{nunk}, V_{pnk} Positive, negative, and bipolar voltages of node k
I_{pj}, I_{nuj}, I_{nj} Net outflow current of the positive, neutral, and negative poles of node j
I_{pk}, I_{nuk}, I_{nk} Net outflow current of the positive, neutral, and negative poles of node k
V_{pnuA}, V_{nunA} Positive and negative port voltages of node A
V_{pnuB},V_{nunB} Positive and negative port voltages of node B
$V_{pA'}$, $V_{nuA'}$,$V_{nA'}$ Positive, neutral and negative node-to-ground voltage of node A
m_p, m_n Transformation ratios of the parallel-type bipolar DC power flow controller
$\boldsymbol{I}_p$, $\boldsymbol{I}_n$, $\boldsymbol{I}_b$ Injection current column vectors of the positive, negative, and bipolar nodes
$\boldsymbol{I}_{Ip}$, $\boldsymbol{I}_{In}$, $\boldsymbol{I}_{Ib}$ Current column vectors provided by the constant current component in positive, negative, and bipolar loads
$\boldsymbol{V}_p$, $\boldsymbol{V}_{nu}$, $\boldsymbol{V}_n$ Voltage N-dimensional column vector of the three networks
$\boldsymbol{V}_{pun}$, $\boldsymbol{V}_{nun}$, $\boldsymbol{V}_{pn}$ Load node port voltage column vectors of positive to neutral, neutral to negative, and positive to negative
$\boldsymbol{I}_{PFCp}$, $\boldsymbol{I}_{PFCn}$ Column vector of the injected current into the network by the equivalent controlled current of the series-type bipolar DC power flow controller connected between the i and j nodes
$\boldsymbol{V}_{b.nodeL}$ Node voltage column vector of the load node
$\boldsymbol{A}_{b.up}$, $\boldsymbol{A}_{b.unu}$, $\boldsymbol{A}_{b.un}$ Branch incidence matrices of each subnetwork
$\boldsymbol{V}_{b.Bp}$, $\boldsymbol{V}_{b.Bnu}$, $\boldsymbol{Vb.Bn}$ Column vectors of branch voltages on the positive, neutral, and negative electrodes
$\boldsymbol{V}_{b.B}$ Branch voltage column vector
$\boldsymbol{V}_{b.node}$ Node-to-ground voltage column vectors of all nodes in the bipolar network

Constants

P_{Ppi} Load power of constant power load of node i
V^0_{pnui} Rated port voltage between positive and neutral poles of node i
Z_{Rpi} Load impedance of constant resistance load of node i
I_{Ipi} Load current of constant current load of node i
V_{es} Voltage of DC electric spring
I_{other} Current of constant current load part of the critical load
R_C Resistance of critical loads
R_N Resistance of noncritical loads
n_p,n_c Transformation ration of the upper and lower DC transformer units
Y_{cp},Y_{cn} Equivalent admittance of the upper and lower units
P_{conv1} Constant power transfer value of the output side of the positive power
P_{conv2} Constant power transfer value of the output side of the negative power
t Turns ratio of the transformer
f_s Operating frequency of the transformer
L Energy storage inductance value of the circuit
D Shift phase ratio
a_{c1}, a_{c2} Power control factor
V_{pc}, V_{nc} Equivalent voltage of the positive and negative poles of the series-type bipolar DC power flow controller
Y_{pAB}, Y_{nuAB}, Y_{nAB} Line admittance of the nodes A and B
L_p, L_n, L_b Positive loads, negative loads and bipolar loads
$\boldsymbol{Y}_{ps}$, $\boldsymbol{Y}_{nus}$, $\boldsymbol{Y}_{ns}$ Connection admittance matrices of the load node and source node
$\boldsymbol{V}_{ps}$, $\boldsymbol{V}_{nus}$, $\boldsymbol{V}_{ns}$ Voltage to ground of the source node in the positive, neutral, and negative networks
$\boldsymbol{Y}_{Rp}$, $\boldsymbol{Y}_{Rn}$, $\boldsymbol{Y}_{Rb}$ Admittance matrices corresponding to the positive, negative, and bipolar constant resistance loads
$\boldsymbol{Y}_p$, $\boldsymbol{Y}_{nu}$, $\boldsymbol{Y}_n$ Node admittance matrices of the three networks after removing the corresponding rows and columns of the source node

The methods in [10–12] improved the accuracy of power flow calculation in the DC distribution network. However, the fastness of power flow calculation cannot be guaranteed because of the iterative algorithm [13].

In [14], the linear power flow calculation method of the DC grid with a converter was established to improve the fastness of power flow calculation of the DC distribution network by deriving the linear expression between node power and voltage deviation. In [15], a Taylor expansion based on linearized power flow model of the DC distribution network was formed. This method avoided multiple iterations of power flow and improved calculation efficiency. References [14,15] improved the speed of power flow calculation of a DC distribution network. However, BDCDN topology was not considered. In [16], a linear power flow calculation method for a DC distribution network under a bipolar structure was derived, but it did not involve the multivoltage-level scenario. In [17], a linearized power flow calculation of multiple-voltage grades BDCDN with control modes of DC transformer was proposed, but it didn't mention the access to other types of flexible equipment. References [18,19] proposed the topology and control method of a DCES and its operation mode under a DC microgrid. However, the power flow algorithm of DCES under BDCDN was not studied. In [20], a power flow calculation method of a unipolar DC distribution network containing DCPFC interconnection was proposed, but the power flow calculation model under a bipolar structure was not derived.

The above references suggest that the existing research has the following limitations: (1) Most of the existing power flow calculation methods of the DC distribution network require multiple iterations to obtain a high-precision convergence solution. However, the rapidity of these calculation methods is limited. (2) The existing linear power flow calculation methods do not comprehensively consider the bipolar network topology and multiple types of flexible equipment with multiple control.

The main contributions of the present work are summarized as follows:

(1) The linearization model of power flow of BDCDN is derived to ensure the accuracy and rapidity of power flow calculation.
(2) The steady-state power flow model for multiple types of flexible equipment is established, considering the influence of multiple control methods. Based on these models, a linear power flow algorithm for BDCDN with multiple types of flexible equipment is proposed, which is verified in 14-bus and 33-bus system and compared with the existing iterative algorithm. This algorithm realizes the rapid power flow analysis of BDCDN.
(3) The influence of different types of flexible equipment on voltage unbalance and voltage deviation is analyzed by using the new proposed algorithm.

2. Structure of BDCDN with multiple types of flexible equipment

Fig. 1 illustrates a BDCDN structure with multiple types of flexible equipment. Flexible equipment consists of interconnection equipment and compensation equipment. Interconnection equipment includes DCTs and DCPFCs. DCTs can transform the electric energy of multiple

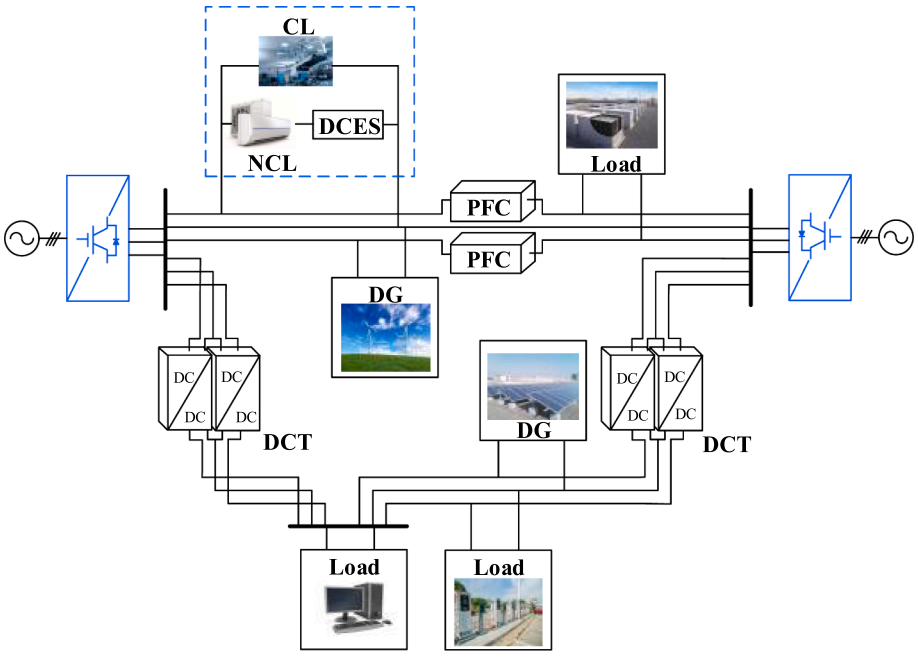


Fig. 1. Network structure of BDCDN.

DC transformation levels [20], control the power flow of public nodes in multiple directions, and ensure the reliable distribution and transfer of power flow during a multi-bus operation. DCPFCs can realize the power flow distribution of one or more lines, reduce the number of transmission lines [21], and improve the power flow flexibility of the ring grid. Compensation equipment includes DCESs. Introducing DCESs in the BDCDN can effectively suppress voltage unbalance. DCESs can ensure that the voltage fluctuations on the critical load (CL) are controlled within the range by transferring the voltage fluctuations to the noncritical load (NCL) [19].

3. Model of load and multiple types of flexible equipment

3.1. ZIP load modeling

The load in the BDCDN can be divided into three types according to the different circuits connected: positive load, negative load, and bipolar load. Each ZIP load consists of constant power load (CPL), constant resistance load (CRL), and constant current load (CCL). The port voltage and load current of the CPL meet the inverse proportional relationship. Thus, the reciprocal of the port voltage is a nonlinear variable. It needs to be converted into a linear variable to reduce the computational complexity caused by nonlinearity. Taylor expansion can be carried out at its operating point when the BDCDN meets the situation in which the voltage of each node exists the upper and lower limits during operation [15].

With the CPL of node 1 as an example, the Taylor expansion of the first two terms is reserved for the reciprocal of the port voltage of the CPL node at the operating point. It can be expressed as

$$I_{Pp1} = P_{Pp1}\left[\frac{2}{V^0_{pnu1}} - \left(\frac{1}{V^0_{pnu1}}\right)^2 V_{pnu1}\right] \tag{1}$$

where V^0_{pnu1} represents the rated voltage at the operating point of node 1, I_{Pp1} represents the load current of CPL of node 1, P_{Pp1} represents the power of CPL of node 1, V_{pnu1} represents the port voltage between positive and neutral poles of node 1. According to (1), CPL can be equivalent to a parallel structure with a CRL and a CCL. Fig. 2(a) shows its equivalent model. As shown in Fig. 2(b), ZIP load can be linearly transformed, and the parameters after transformation can be expressed as

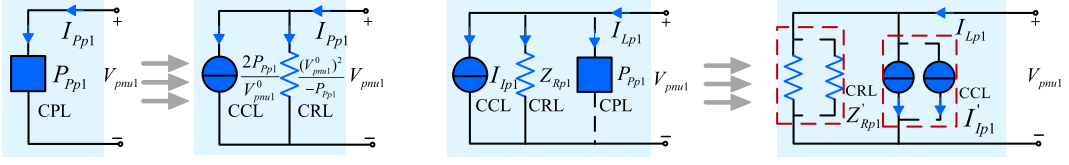
$$\begin{cases} Z'_{Rp1} = Z_{Rp1} // \dfrac{-(V^0_{pnu1})^2}{P_{Pp1}} \\ I'_{Ip1} = I_{Ip1} + \dfrac{2P_{Pp1}}{V^0_{pnu1}} \end{cases} \tag{2}$$

where Z_{Rp1} represents the resistance of CRL of node1, I_{Ip1} represents the load current of CCL of node 1.

3.2. DCES modeling

The DCES can realize the automatic balance of power and the voltage stabilization of the CL.

The equivalent model of the DCES when it reaches the stable state is



(a) CPL model;(b) ZIP model

Fig.2. Equivalent model of linearized CPL.

shown in Fig. 3(a). V is the port voltage of the CL and smart load branches. V_{es} is the output voltage of the DCES. The output voltage of the DCES is constant under stable conditions. Thus, the DCES can be equivalent to a voltage-controlled source with a certain value. NCL is specified as CRL, and CL can be different combinations of CPL, CCL, and CRL.

The relationship between injection current and voltage is expressed as

$$I = \frac{V}{R_C} + \frac{V - V_{es}}{R_N} + I_{other} = V\left(\frac{1}{R_C} + \frac{1}{R_N}\right) - \frac{V_{es}}{R_N} + I_{other} \tag{3}$$

where R_N represents the NCL's resistance, R_C represents the CRL part of the CL, and I_{other} represents the current of CCL part of the CL. According to (3), the DCES can also be regarded as a parallel combination of a CCL and a CRL, as shown in Fig. 3(b), where $(1/R_C + 1/R_N)$ represents the resistance value of the CRL, and $(I_{other}\text{-}V_{es}/R_N)$ represents the current value of the CCL. Therefore, the node containing DCES can be considered connected to a parallel combination of a CRL and a CCL.

3.3. Bipolar DC Transformer(BDCT) modeling

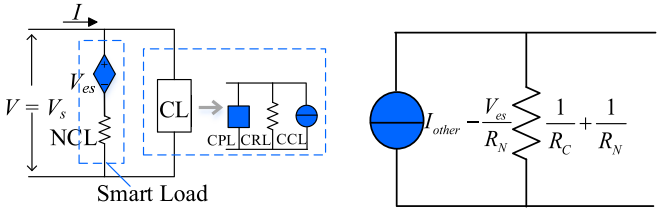
Many topological structures of DCTs, such as non-isolated buck/boost type and dual active bridge (DAB), are suitable for a distribution network. Different topologies have differences in control methods. However, the conversion of DC voltage level and the function of voltage and power transmission can be both realized in the steady state [22].

DCT control strategies are divided into three types: constant transmission ratio control (CRC), constant power control (CPC), and constant voltage control (CVC) [23]. CRC represents a control strategy that does not deliberately adjust the control signal and realize voltage level conversion with a fixed transformation ratio [24]. CPC represents a control strategy that adjusts control signals according to certain principles. It makes the DCT transmit fixed power through open-loop control. CVC means a control strategy that adjusts control signal by using closed-loop control. Thus, the voltage on the side of the DCT is the set value.

The schematic diagram of a BDCT, shown in Fig. 4, can be inferred according to the DC–DC converter topology in a bipolar high-voltage DC transmission line [25]. In particular, a DC–DC converter unit is symmetrically connected between the positive pole and the neutral line and between the negative pole and the neutral line. V_{pnuj}, V_{nunj}, V_{pnj}, V_{pnuk}, V_{nunk}, and V_{pnk} represent the positive, negative, and bipolar voltages of node j and k. I_{pj}, I_{nuj}, I_{nj}, I_{pk}, I_{nuk}, and I_{nk} represent the net outflow current of the positive, neutral, and negative poles of node j and k. n_p and n_n represent the transformation ratio of the upper and lower DCT units, respectively. Y_{cp} and Y_{cn} represent the equivalent admittance of the upper and lower units reduced to the output side. The parameters of the two DCTs are the same. They can be obtained using $Y_c = Y_{cp} = Y_{cn}$ and $n_c = n_p = n_n$ in the case of constant transformation ratio.

(1) CRC strategy.

Under the CRC strategy, the node current equation of the nodes j and k is written according to the transformer characteristics. The linear



(a) DCES model; (b) CCL and CRL combination.

Fig. 3. Model of DCES.

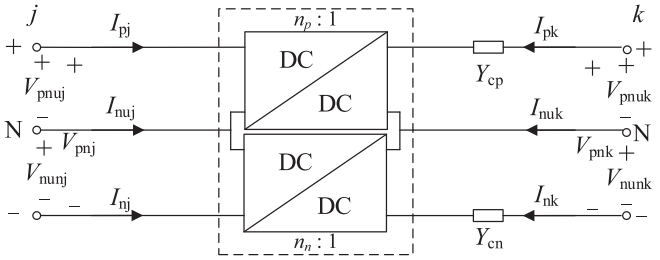


Fig. 4. Schematic diagram of BDCT.

transformation is performed by subtracting the node equations two by two. Then, the voltage and current transfer model of the high- and low-voltage sides of the BDCT can be obtained under this strategy. It can be expressed as

$$\begin{bmatrix} I_{pj} - I_{nuj} \\ I_{pk} - I_{nuk} \\ I_{nuj} - I_{nj} \\ I_{nuk} - I_{nk} \\ I_{pj} - I_{nj} \\ I_{pk} - I_{nk} \end{bmatrix} = \begin{bmatrix} \frac{2Y_c}{n_c^2} & \frac{-2Y_c}{n_c} & \frac{-Y_c}{n_c^2} & \frac{Y_c}{n_c} & 0 & 0 \\ \frac{-2Y_c}{n_c} & 2Y_c & \frac{Y_c}{n_c} & -Y_c & 0 & 0 \\ \frac{-Y_c}{n_c^2} & \frac{Y_c}{n_c} & \frac{2Y_c}{n_c^2} & \frac{-2Y_c}{n_c} & 0 & 0 \\ \frac{Y_c}{n_c} & -Y_c & \frac{-2Y_c}{n_c^2} & 2Y_c & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{Y_c}{n_c^2} & \frac{-Y_c}{n_c} \\ 0 & 0 & 0 & 0 & \frac{-Y_c}{n_c} & Y_c \end{bmatrix} \begin{bmatrix} V_{pnuj} \\ V_{pnuk} \\ V_{nunj} \\ V_{nunk} \\ V_{pnj} \\ V_{pnk} \end{bmatrix} \tag{4}$$

(2) CPC strategy.

Under the CPC strategy, one side of the BDCT exhibits CPL characteristics, and the other side exhibits CPS characteristics. The voltage and current relationship of the positive and negative pole of the secondary side of the DCT is listed according to the characteristics of CPC. It can be expressed as

$$\begin{cases} V_{pnuk} I_{pk} = -P_{conv1} \\ V_{nunk} I_{nk} = P_{conv2} \end{cases} \tag{5}$$

where the P_{conv1} represents the constant power transfer value of the output side of the positive power under the CPC, and P_{conv2} represents the constant power transfer value of the output side of the negative power. The relationship equations between the voltage and current on both sides of the BDCT can be listed according to the equivalent model and the transformation characteristics of the transformer. They can be expressed as

$$\begin{cases} V_{pnuj} = n_p\left(\frac{-I_{pk}}{Y_{cp}} + V_{pnuk}\right) \\ V_{nunj} = n_n\left(\frac{I_{nk}}{Y_{cn}} + V_{nunk}\right) \\ I_{pk} = -n_p I_{pj} \\ I_{nk} = -n_n I_{nj} \end{cases} \tag{6}$$

Equations (5) and (6) are combined to obtain the expressions of I_{pj}, I_{nj}, I_{nuj}, I_{pk}, I_{nk}, and I_{nuk}, they can be expressed as.

$$\begin{cases} I_{pj} = \dfrac{P_{conv1}^2}{V_{pnuj}V_{pnuk}^2Y_{cp}} + \dfrac{P_{conv1}}{V_{pnuj}} \\ I_{nuj} = \dfrac{P_{conv2}^2}{V_{pnuj}V_{pnuk}^2Y_{cp}} + \dfrac{P_{conv2}}{V_{pnuj}} - \dfrac{P_{conv1}^2}{V_{pnuj}V_{pnuk}^2Y_{cp}} - \dfrac{P_{conv1}}{V_{pnuj}} \\ I_{nj} = \dfrac{-P_{conv2}^2}{V_{pnuj}V_{pnuk}^2Y_{cp}} - \dfrac{P_{conv2}}{V_{pnuj}} \\ I_{pk} = \dfrac{-P_{conv1}}{V_{pnuk}} \\ I_{nuk} = \dfrac{P_{conv1}}{V_{pnuk}} - \dfrac{P_{conv2}}{V_{nunk}} \\ I_{nk} = \dfrac{-P_{conv2}}{V_{nunk}} \end{cases} \tag{7}$$

For an isolated BDCT with the commonly used CPC strategy, the power value transmitted to the output side in a steady state is a function of the primary- and secondary-side voltage [26]. The DAB-type BDCT under the single-phase-shift control strategy is taken as an example. In this case, the power value transmitted to the output side is shown in (8), where t is the turns ratio of the transformer, f_s is the operating frequency of the transformer, L is the energy storage inductance value of the circuit, and D is the shift phase ratio. DAB transformer realizes CPC strategy under the condition that the voltage on both sides does not change much by changing D. The transmission power of the BDCT can be replaced by (9) when the influence of the BDCT CPC strategy on the power flow of the DC distribution network is considered. Moreover, the influence of the CPC method can be converted into the power control factor a_{c1} and a_{c2} of the upper and lower two unipolar BDCT units, thereby simplifying the transfer equation.

$$P_{\text{conv}} = \frac{tD(1-D)}{2f_sL}V_jV_k \tag{8}$$

$$\begin{cases} P_{\text{conv1}} = a_{c1}V_{pnuj}V_{pnuk} \\ P_{\text{conv2}} = a_{c2}V_{nunj}V_{nunk} \end{cases} \tag{9}$$

Equation (9) is incorporated into (7). Then, the linear transformation of two-by-two subtraction on the node equations of the three poles of the j and k nodes is performed. The voltage and current transfer model of the high and low sides of the BDCT under the CPC strategy is obtained. It can be expressed as

$$\begin{bmatrix} I_{pj}-I_{nuj} \\ I_{pk}-I_{nuk} \\ I_{nuj}-I_{nj} \\ I_{nuk}-I_{nk} \\ I_{pj}-I_{nj} \\ I_{pk}-I_{nk} \end{bmatrix} = \begin{bmatrix} \frac{2a_{c1}^2}{Y_c} & 2a_{c1} & \frac{-a_{c2}^2}{Y_c} & -a_{c2} & 0 & 0 \\ -2a_{c1} & 0 & a_{c2} & 0 & 0 & 0 \\ \frac{-a_{c1}^2}{Y_c} & -a_{c1} & \frac{2a_{c2}^2}{Y_c} & 2a_{c2} & 0 & 0 \\ a_{c1} & 0 & -2a_{c2} & 0 & 0 & 0 \\ \frac{a_{c1}^2-a_{c2}^2}{Y_c} & a_{c1}-a_{c2} & 0 & 0 & \frac{a_{c2}^2}{Y_c} & a_{c2} \\ -a_{c1}+a_{c2} & 0 & 0 & 0 & -a_{c2} & 0 \end{bmatrix} \begin{bmatrix} V_{pnuj} \\ V_{pnuk} \\ V_{nunj} \\ V_{nunk} \\ V_{pnj} \\ V_{pnk} \end{bmatrix} \tag{10}$$

(3) CVC strategy.

Under the CVC strategy, the power input side of the BDCT can be regarded as two CPLs connected to the positive and negative terminals. The power output side can be regarded as two constant source loads (CSLs) connected to the different terminals. In the power flow calculation, the decoupling calculation of the networks on both sides of the BDCT can be performed. In particular, the calculation of the CVC is performed first, and the output current of the unit transformer in the BDCT is obtained. The equivalent power of the input side of the BDCT is obtained by power equations. Then, it is substituted into the power flow calculation model as a CPL to complete the calculation. The power equations of the BDCT under the CVC strategy can be expressed as

$$\begin{cases} P_{\text{c1 - in}} = P_{\text{c1}} + P_{\text{loss1}} = -V_{pnuk}I_{pk} + \dfrac{I_{pk}^2}{Y_c} \\ P_{\text{c2 - in}} = P_{\text{c2}} + P_{\text{loss2}} = -V_{nnuk}I_{nk} + \dfrac{I_{nk}^2}{Y_c} \end{cases} \tag{11}$$

where $P_{\text{c1-in}}$ and $P_{\text{c2-in}}$ can represent the equivalent power of the upper- and lower-unit transformer power input sides, P_{c1} and P_{c2} represent the power of the upper- and lower-unit transformer power output sides, respectively, and P_{loss1} and P_{loss2} represent the power loss of the upper- and lower-unit transformer, respectively.

3.4. DCPFC modeling

To realize the effective regulation of the internal power flow of the entire DC power grid, the DC system usually needs to install a DCPFC to improve the line power flow distribution to ensure the safe, stable and economical operation of the system [27]. In a DC distribution network, the voltage of each node only involves amplitude, and the line impedance only involves resistance. Therefore, the power adjustment in the DC system can only be realized by changing the node voltage amplitude and the line resistance. Hence, DCPFC can be divided into two types: resistance type and voltage type [27]. Voltage-type DCPFC have obvious advantages in power flow regulation range and system additional loss [27]. Thus, this study takes the voltage-type DCPFC as the research focus.

The voltage-type unipolar DCPFC can be divided into parallel and series-type DCPFCs [27]. In the bipolar wiring mode, the connection modes of parallel-type and series-type DCPFC are shown in Fig. 5. Among them, V_{pnuA}, V_{nunA}, V_{pnuB} and V_{nunB} respectively represent the positive and negative port voltages of nodes A and B, $V_{\text{pA'}}$, $V_{\text{nuA'}}$ and $V_{\text{nA'}}$ respectively represent the positive, neutral and negative node-to-ground voltage, m_p, m_n represent the transformation ratios of the two converter units of the parallel-type bipolar DCPFC, V_{pc}, V_{nc} represent the equivalent voltage of the positive and negative poles of the series-type bipolar DCPFC, Y_{pAB}, Y_{nuAB}, Y_{nAB} respectively represent the line admittance of the nodes A and B.

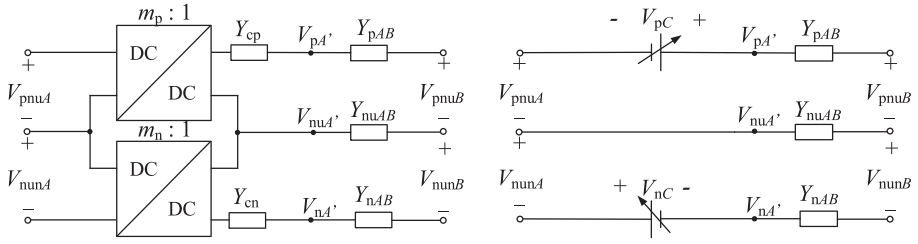
The topology of the parallel-type bipolar DCPFC is the same as the bipolar DCT under CRC, so its voltage and current transfer function of high and low voltage side is the same as that of bipolar DCT, the details are as shown in section 3.3.

Due to the existence of the neutral resistance, the voltage and current transfer characteristics of the positive and negative port voltage of the series-type bipolar DCPFC are coupled, so it can not be simply equivalent to the CCL connected to the positive and negative terminals of the front and rear nodes. According to the Thevenin/Norton theorem, the structure of a voltage source connected with a resistor in series can be equivalent to the structure of a current source connected with a resistor in parallel. Therefore, the model of series-type bipolar DCPFC can be equivalent to the combination of the positive, negative line resistances and the controlled current sources whose parameters are $V_{pC}Y_{pAB}$ and $V_{nC}Y_{nAB}$, as shown in Fig. 6.

4. Linear power flow calculation of BDCDN with multiple types of flexible equipment

4.1. Node equations

The basic structure of BDCDN with flexible equipment is shown in Fig. 7. The network contains three lines, namely positive, neutral and negative line, which are denoted by “p”, “nu” and “n” respectively. All nodes are divided into source nodes and load nodes. There are N load nodes and M source nodes. V_{pi}, V_{nui} and V_{ni} respectively represent the voltages of positive, neutral and negative nodes of the i node relative to



(a) parallel-type bipolar DCPFC; (b) series-type bipolar DCPFC

Fig. 5. Equivalent model of bipolar voltage-based DCPFC.

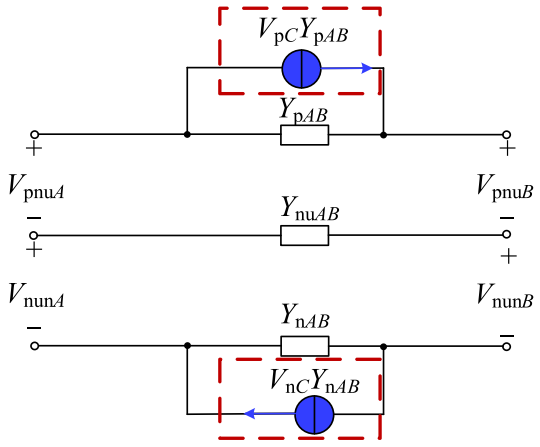


Fig. 6. Equivalent current source model of the series-type bipolar DCPFC.

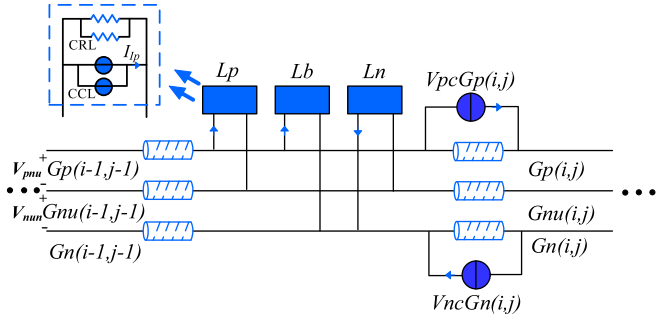


Fig. 7. Structure of BDCDN with multiple types of flexible equipment.

the ground, $G_{p(i,j)}$, $G_{nu(i,j)}$, $G_{n(i,j)}$ represent the line admittance between nodes i and j on the positive, neutral and negative line, respectively. The loads in the BDCN are divided into three types according to the connected lines: positive load, negative load and bipolar load, which are represented by L_p, L_n and L_b respectively. It can be seen from Section 3.1 that each load contains three loads with different characteristics, namely CRL, CCL and CPL.

To analyze the network structure of BDCDN with flexible equipment, the loads connected to the line are regarded ae current sources, the DCES is equivalent to the parallel combination of CCL and CRL, and the DCPFC is equivalent to the combination of the resistances of positive and negative lines and the controlled current source. Positive poles, neutral lines, and negative poles are decoupled, and the node equations on the three lines of the three poles are listed for N load nodes. They can be expressed as

$$\boldsymbol{I}_{\mathrm{p}} = -\boldsymbol{I}_{\mathrm{lp}} - \boldsymbol{I}_{\mathrm{lb}} + \boldsymbol{Y}_{\mathrm{ps}}\boldsymbol{V}_{\mathrm{ps}} + \boldsymbol{I}_{\mathrm{PFCp}} - \boldsymbol{Y}_{\mathrm{Rp}}\boldsymbol{V}_{\mathrm{pnu}} - \boldsymbol{Y}_{\mathrm{Rb}}\boldsymbol{V}_{\mathrm{pn}} = \boldsymbol{Y}_{\mathrm{p}}\boldsymbol{V}_{\mathrm{p}} \tag{12}$$

$$\boldsymbol{I}_{\mathrm{nu}} = \boldsymbol{I}_{\mathrm{lp}} - \boldsymbol{I}_{\mathrm{ln}} + \boldsymbol{Y}_{\mathrm{Rp}}\boldsymbol{V}_{\mathrm{pnu}} - \boldsymbol{Y}_{\mathrm{Rn}}\boldsymbol{V}_{\mathrm{nun}} = \boldsymbol{Y}_{\mathrm{nu}}\boldsymbol{V}_{\mathrm{nu}} \tag{13}$$

$$\boldsymbol{I}_{\mathrm{n}} = \boldsymbol{I}_{\mathrm{ln}} + \boldsymbol{I}_{\mathrm{lb}} + \boldsymbol{Y}_{\mathrm{ns}}\boldsymbol{V}_{\mathrm{ns}} + \boldsymbol{I}_{\mathrm{PFCn}} + \boldsymbol{Y}_{\mathrm{Rn}}\boldsymbol{V}_{\mathrm{nun}} + \boldsymbol{Y}_{\mathrm{Rb}}\boldsymbol{V}_{\mathrm{pn}} = \boldsymbol{Y}_{\mathrm{n}}\boldsymbol{V}_{\mathrm{n}} \tag{14}$$

$\boldsymbol{I}_p$, $\boldsymbol{I}_n$, and $\boldsymbol{I}_b$ are the N-dimensional injection current column vectors of the positive, negative, and bipolar nodes, respectively. $\boldsymbol{I}_{lp}$, $\boldsymbol{I}_{ln}$, and $\boldsymbol{I}_{lb}$ are the N-dimensional current column vectors provided by the constant current component in positive, negative, and bipolar loads, respectively. The constant current component includes the equivalent constant current component of the node containing DCES and the equivalent constant current component of the load node containing only different load combinations. $\boldsymbol{Y}_{ps}$, $\boldsymbol{Y}_{nus}$, and $\boldsymbol{Y}_{ns}$ are the connection admittance matrices of the load node and source node. Their order is $N \times M$. The internal element $\boldsymbol{Y}_{psij}$ represents the connection admittance value between the i-th load node and the j-th source node. The three poles generally use the same cable line, and the network structures are symmetrical. Thus, the line resistances on the three lines can be regarded as equal, that is, $\boldsymbol{Y}_{ps} = \boldsymbol{Y}_{nus} = \boldsymbol{Y}_{ns}$. $\boldsymbol{V}_{ps}$, $\boldsymbol{V}_{nus}$, and $\boldsymbol{V}_{ns}$ represent the voltage to ground of the source node in the positive, neutral, and negative networks. They are all M dimensional column vectors because the source is symmetrical, that is, $\boldsymbol{V}_{ps} = -\boldsymbol{V}_{ns}$, $\boldsymbol{V}_{nus} = 0$. $\boldsymbol{Y}_{Rp}$, $\boldsymbol{Y}_{Rn}$, and $\boldsymbol{Y}_{Rb}$ are the admittance matrices corresponding to the positive, negative, and bipolar CRLs. The CRL situation also contains the equivalent CRL of the node containing DCES or DCPFC. The order of these admittance matrices is $N \times N$, and they only have elements on the diagonal. Given $\boldsymbol{Y}_{Rp}$ as an example, the element $\boldsymbol{Y}_{Rp(i,i)}$ in the i-th row and i-th column is equal to $1/\boldsymbol{Z}_{Rpi}$. $\boldsymbol{Y}_p$, $\boldsymbol{Y}_{nu}$, and $\boldsymbol{Y}_n$ are the node admittance matrices of the three networks after removing the corresponding rows and columns of the source node, and their order is $N \times N$. Given the symmetry of the line, the node admittance matrices can also be regarded as $\boldsymbol{Y}_p = \boldsymbol{Y}_{nu} = \boldsymbol{Y}_n = \boldsymbol{Y}_L$. $\boldsymbol{V}_p$, $\boldsymbol{V}_{nu}$, and $\boldsymbol{V}_n$ represent the voltage N-dimensional column vector of the three networks, reflecting the voltage of each node with respect to ground. $\boldsymbol{V}_{pun}$, $\boldsymbol{V}_{nun}$, and $\boldsymbol{V}_{pn}$ represent the load node port voltage column vectors of positive to neutral, neutral to negative, and positive to negative, and they are all N-dimensional column vectors. They can be regarded as $\boldsymbol{V}_{pun} = \boldsymbol{V}_p - \boldsymbol{V}_{nu}$, $\boldsymbol{V}_{nun} = \boldsymbol{V}_{nu} - \boldsymbol{V}_n$, and $\boldsymbol{V}_{pn} = \boldsymbol{V}_p - \boldsymbol{V}_n$. $\boldsymbol{I}_{\mathrm{PFCp}}$, $\boldsymbol{I}_{\mathrm{PFCn}}$ represent the N-dimensional column vector of the injected current into the network by the equivalent controlled current of the series-type bipolar DCPFC connected between the i and j nodes. And they can be expressed as

$$\begin{cases} \boldsymbol{I}_{\mathrm{PFCp}} = [\overbrace{0 \quad \cdots \quad \underset{nodei}{-V_{\mathrm{pC}}G_{\mathrm{pij}}} \quad \cdots \quad \underset{nodej}{V_{\mathrm{pC}}G_{\mathrm{pij}}} \quad \cdots \quad 0}^{Nnode}]^{\mathrm{T}} \\ \boldsymbol{I}_{\mathrm{PFCn}} = [\overbrace{0 \quad \cdots \quad \underset{nodei}{V_{\mathrm{nC}}G_{\mathrm{nij}}} \quad \cdots \quad \underset{nodej}{-V_{\mathrm{nC}}G_{\mathrm{nij}}} \quad \cdots \quad 0}^{Nnode}]^{\mathrm{T}} \end{cases} \tag{15}$$

4.2. Voltage calculation model

According to Eqs. (12)–(14), the linear transformation of subtracting two by two is performed. Moreover, the relationship between the port

voltage and the load current is obtained, as shown in Eq.(16). The related variable expression is shown in Eq.(17). The linear power flow model expression of BDCDN with multiple types of flexible equipment can be obtained through Eq.(16), as shown in Eq.(18). And the variable E represents the identity matrix of order N by N.

$$\boldsymbol{M}\boldsymbol{I}_{\mathrm{LI}}+\boldsymbol{I}_{\mathrm{b.s}}+\boldsymbol{I}_{\mathrm{b.PFC}}=\boldsymbol{Y}_{\mathrm{b.L}}\boldsymbol{V}_{\mathrm{b.L}} \tag{16}$$

$$\begin{cases} \boldsymbol{M}=\begin{bmatrix} -2\boldsymbol{E} & \boldsymbol{E} & -\boldsymbol{E} \\ \boldsymbol{E} & -2\boldsymbol{E} & -\boldsymbol{E} \\ -\boldsymbol{E} & -\boldsymbol{E} & -2\boldsymbol{E} \end{bmatrix} \\ \boldsymbol{I}_{\mathrm{LI}}=[\boldsymbol{I}_{\mathrm{Ip}} \quad \boldsymbol{I}_{\mathrm{In}} \quad \boldsymbol{I}_{\mathrm{Ib}}]^{\mathrm{T}} \\ \boldsymbol{I}_{\mathrm{b.s}}=\left[\boldsymbol{Y}_{\mathrm{ps}}\boldsymbol{V}_{\mathrm{ps}} \quad -\boldsymbol{Y}_{\mathrm{ns}}\boldsymbol{V}_{\mathrm{ns}} \quad \left(\boldsymbol{Y}_{\mathrm{ps}}\boldsymbol{V}_{\mathrm{ps}}-\boldsymbol{Y}_{\mathrm{ns}}\boldsymbol{V}_{\mathrm{ns}}\right)\right]^{\mathrm{T}} \\ \boldsymbol{Y}_{\mathrm{b.L}}=\begin{bmatrix} \boldsymbol{Y}_{\mathrm{L}}+2\boldsymbol{Y}_{\mathrm{Rp}} & -\boldsymbol{Y}_{\mathrm{Rn}} & \boldsymbol{Y}_{\mathrm{Rb}} \\ -\boldsymbol{Y}_{\mathrm{Rp}} & \boldsymbol{Y}_{\mathrm{L}}+2\boldsymbol{Y}_{\mathrm{Rn}} & \boldsymbol{Y}_{\mathrm{Rb}} \\ \boldsymbol{Y}_{\mathrm{Rp}} & \boldsymbol{Y}_{\mathrm{Rn}} & \boldsymbol{Y}_{\mathrm{L}}+2\boldsymbol{Y}_{\mathrm{Rb}} \end{bmatrix} \\ \boldsymbol{V}_{\mathrm{b.L}}=[\boldsymbol{V}_{\mathrm{pnu}} \quad \boldsymbol{V}_{\mathrm{nun}} \quad \boldsymbol{V}_{\mathrm{pn}}]^{\mathrm{T}} \\ \boldsymbol{I}_{\mathrm{b.PFC}}=[\boldsymbol{I}_{\mathrm{PFCp}} \quad -\boldsymbol{I}_{\mathrm{PFCn}} \quad \left(\boldsymbol{I}_{\mathrm{PFCp}}-\boldsymbol{I}_{\mathrm{PFCn}}\right)]^{\mathrm{T}} \end{cases} \tag{17}$$

$$\boldsymbol{V}_{\mathrm{b.L}}=\boldsymbol{Y}_{\mathrm{b.L}}^{-1}(\boldsymbol{M}\boldsymbol{I}_{\mathrm{LI}}+\boldsymbol{I}_{\mathrm{b.s}}+\boldsymbol{I}_{\mathrm{b.PFC}}) \tag{18}$$

4.3. Branch current calculation model

The node voltage column vector $\boldsymbol{V}_{\mathrm{b.nodeL}}$ of the load node is calculated by using the solved voltage results, combined with the node equations of the bipolar network and Eqs.(12) to (14). It is shown in Eq. (19).

$$\begin{aligned} \boldsymbol{V}_{\mathrm{b.nodeL}} &= [\boldsymbol{V}_{\mathrm{p}} \quad \boldsymbol{V}_{\mathrm{nu}} \quad \boldsymbol{V}_{\mathrm{n}}]^{\mathrm{T}} \\ &= \begin{bmatrix} \boldsymbol{Y}_{\mathrm{L}} & 0 & 0 \\ 0 & \boldsymbol{Y}_{\mathrm{L}} & 0 \\ 0 & 0 & \boldsymbol{Y}_{\mathrm{L}} \end{bmatrix}^{-1} \left\{ \begin{bmatrix} -\boldsymbol{E} & 0 & -\boldsymbol{E} \\ \boldsymbol{E} & -\boldsymbol{E} & 0 \\ 0 & \boldsymbol{E} & \boldsymbol{E} \end{bmatrix} \boldsymbol{I}_{\mathrm{LI}} + \begin{bmatrix} \boldsymbol{Y}_{\mathrm{ps}}\boldsymbol{V}_{\mathrm{ps}} \\ 0 \\ \boldsymbol{Y}_{\mathrm{ns}}\boldsymbol{V}_{\mathrm{ns}} \end{bmatrix} \right. \\ &\quad \left. + \begin{bmatrix} \boldsymbol{I}_{\mathrm{PFCp}} \\ 0 \\ \boldsymbol{I}_{\mathrm{PFCn}} \end{bmatrix} + \begin{bmatrix} -\boldsymbol{Y}_{\mathrm{Rp}} & 0 & -\boldsymbol{Y}_{\mathrm{Rb}} \\ \boldsymbol{Y}_{\mathrm{Rp}} & -\boldsymbol{Y}_{\mathrm{Rn}} & 0 \\ 0 & \boldsymbol{Y}_{\mathrm{Rn}} & \boldsymbol{Y}_{\mathrm{Rb}} \end{bmatrix} \boldsymbol{V}_{\mathrm{b.L}} \right\} \end{aligned} \tag{19}$$

The branches are numbered to form the branch incidence matrices $\boldsymbol{A}_{\mathrm{b.up}}$, $\boldsymbol{A}_{\mathrm{b.unu}}$, and $\boldsymbol{A}_{\mathrm{b.un}}$ of each subnetwork. The branch distribution on each line is the same. The branch incidence matrix satisfies $\boldsymbol{A}_{\mathrm{b.up}}=\boldsymbol{A}_{\mathrm{b.unu}}=\boldsymbol{A}_{\mathrm{b.un}}$. The branch incidence matrix $\boldsymbol{A}_{\mathrm{b.u}}$ of the bipolar network is formed according to the branch incidence matrix on the line. Its expression is $\boldsymbol{A}_{\mathrm{b.u}}=\mathrm{diag}(\boldsymbol{A}_{\mathrm{b.up}}\boldsymbol{A}_{\mathrm{b.unu}}.\boldsymbol{A}_{\mathrm{b.un}})$. The branch voltage column vector $\boldsymbol{V}_{\mathrm{b.B}}$ of the bipolar network is obtained using Eq. (20) and the relationship of branch voltage and node voltage.

$$\boldsymbol{V}_{\mathrm{b.B}}=\left[\boldsymbol{V}_{\mathrm{b.Bp}} \quad \boldsymbol{V}_{\mathrm{b.Bnu}} \quad \boldsymbol{V}_{\mathrm{b.Bn}}\right]^{\mathrm{T}}=\boldsymbol{A}_{\mathrm{b.u}}^{\mathrm{T}}\boldsymbol{V}_{\mathrm{b.node}} \tag{20}$$

In the formula, $\boldsymbol{V}_{\mathrm{b.Bp}}$, $\boldsymbol{V}_{\mathrm{b.Bnu}}$, and $\boldsymbol{V}_{\mathrm{b.Bn}}$ represent the column vectors of branch currents on the positive, neutral, and negative electrodes, respectively. $\boldsymbol{V}_{\mathrm{b.node}}$ represents the node-to-ground voltage column vectors of all nodes in the bipolar network, that is, the column vector adds the node voltage value of the source node to the $\boldsymbol{V}_{\mathrm{b.nodeL}}$.

The bipolar network branch admittance matrix $Y_{\mathrm{b.B}}$ is written according to the branch impedance on each line. The calculation model of the branch current column vector $I_{\mathrm{b.B}}$ is obtained, and correct the branch current of the branch where the series-type bipolar DCPFC is located. They can be expressed as

$$\boldsymbol{I}_{\mathrm{b.B}}=[\boldsymbol{I}_{\mathrm{b.Bp}} \quad \boldsymbol{I}_{\mathrm{b.Bnu}} \quad \boldsymbol{I}_{\mathrm{b.Bn}}]^{\mathrm{T}}=\boldsymbol{Y}_{\mathrm{b.B}}\boldsymbol{V}_{\mathrm{b.B}} \tag{21}$$

$$\begin{cases} \boldsymbol{I}'_{\mathrm{b.Bpij}}=\boldsymbol{I}_{\mathrm{b.Bpij}}+\boldsymbol{V}_{\mathrm{pC}}\boldsymbol{Y}_{\mathrm{pij}} \\ \boldsymbol{I}'_{\mathrm{b.Bnuij}}=\boldsymbol{I}_{\mathrm{b.Bnuij}} \\ \boldsymbol{I}'_{\mathrm{b.Bnij}}=\boldsymbol{I}_{\mathrm{b.Bnij}}-\boldsymbol{V}_{\mathrm{nC}}\boldsymbol{Y}_{\mathrm{nij}} \end{cases} \tag{22}$$

4.4. Power flow calculation process

(1) Power flow calculation without CVC.

The Section 3.3 reveals that under the CRC and CPC strategies, the influence of adding the BDCT on the power flow calculation is only to modify the admittance matrix. And after adding the DCES, only need to change the N-dimension current column vector provided by the constant current components in different loads and the admittance matrix corresponding to CRL of three poles. Thus, the power flow calculation can be performed according to the steps described below.

Step 1: Assign the initial value of each node voltage, which is the rated voltage value.
Step 2: According to the relevant parameters of the distribution network, convert the CPLs connected to the load node without DCES into the CCLs and CRLs. Then, transform the node containing DCES into CCL and CRL. Thus, it can form matrices of $\boldsymbol{Y}_{Rp}$, $\boldsymbol{Y}_{Rn}$, and $\boldsymbol{Y}_{Rb}$ and vector $\boldsymbol{I}_{LI}$. Then, write matrix $\boldsymbol{Y}_{b.s}$ of each source node and load node and combine it with matrix $\boldsymbol{Y}_{b.L}$. The transfer model of DCT in CRC and CVC, namely, (4) and (10), modifies $\boldsymbol{Y}_{b.L}$.
Step 3: Substitute each matrix into (14) to calculate the node port voltage column vector $\boldsymbol{V}_{b.L}$ of the load node of the bipolar network.
Step 4: The primary and secondary sides of the BDCT are isolated from each other, and the zero potential points are inconsistent. Thus, node-to-ground voltage of the primary and secondary sides must be calculated independently. Therefore, the networks must be decoupled, and the neutral grounded outlet node of the grounded BDCT must be made a source node of the secondary side (the neutral potential is zero). The other inlet and outlet nodes of the BDCT are all regarded as load nodes. The influence of the BDCT is converted into the injection of the current source at the inlet and outlet nodes. For

the secondary side network, (4) and (10) are used to calculate the net outflow current at the inlet and outlet nodes. (23) is used to calculate the node voltage of the three lines. (21) is used to calculate the current of each branch.

$$\begin{cases} P_{pij} = V_{pi} I_{pij} \\ P_{pji} = V_{pj} I_{pji} \\ \Delta P_{pij} = P_{pij} + P_{pji} \end{cases} \tag{24}$$

$$\boldsymbol{V}_{\text{b.nodeL}} = [\boldsymbol{V}_{pL} \quad \boldsymbol{V}_{nuL} \quad \boldsymbol{V}_{nL}]^T = \begin{bmatrix} \boldsymbol{Y}_L & 0 & 0 \\ 0 & \boldsymbol{Y}_L & 0 \\ 0 & 0 & \boldsymbol{Y}_L \end{bmatrix}^{-1} \left\{ \begin{bmatrix} -\boldsymbol{E} & 0 & -\boldsymbol{E} \\ \boldsymbol{E} & -\boldsymbol{E} & 0 \\ 0 & \boldsymbol{E} & \boldsymbol{E} \end{bmatrix} \boldsymbol{I}_{IL} + \begin{bmatrix} \boldsymbol{Y}_{ps} \boldsymbol{V}_{ps} \\ 0 \\ \boldsymbol{Y}_{ns} \boldsymbol{V}_{ns} \end{bmatrix} + \begin{bmatrix} \boldsymbol{I}_{\text{PFCp}} \\ 0 \\ \boldsymbol{I}_{\text{PFCn}} \end{bmatrix} + \begin{bmatrix} -\boldsymbol{Y}_{Rp} & 0 & -\boldsymbol{Y}_{Rb} \\ \boldsymbol{Y}_{Rp} & -\boldsymbol{Y}_{Rn} & 0 \\ 0 & \boldsymbol{Y}_{Rn} & \boldsymbol{Y}_{Rb} \end{bmatrix} \boldsymbol{V}_{b.L} - \begin{bmatrix} \boldsymbol{I}_{pc} \\ \boldsymbol{I}_{nuc} \\ \boldsymbol{I}_{nc} \end{bmatrix} \right\} \tag{23}$$

Step 5: For the primary side network, the method in step 4 is used to calculate the current of each branch of the primary side network.

Step 6: Calculate the line power and line loss on the three lines according to the calculation results of the node-to-ground voltage and branch current. Then, complete the power flow calculation. With the positive line as an example, the calculation formula can be expressed as

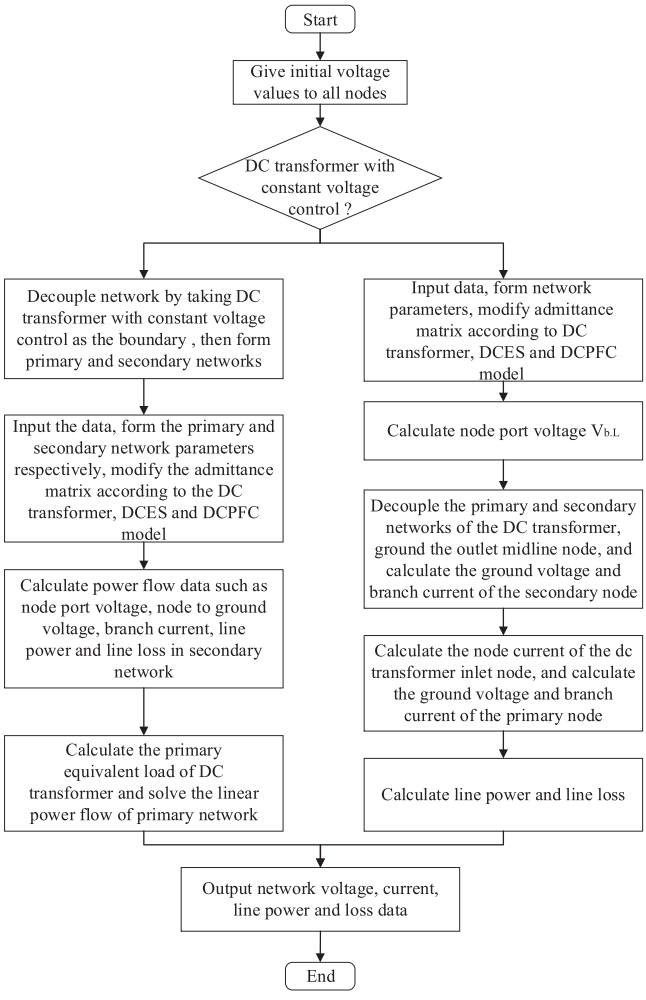


Fig. 8. Flow chart of power flow calculation.

(2) Power flow calculation with CVC.

The previous analysis reveals that the network on both sides of the BDCT under the CVC strategy can be decoupled. Thus, the power flow calculation can be performed as follows.

Step 1: Decouple the networks on both sides with the CVC transformer as the boundary. Treat the transformer inlet node as the load node of the primary network and the outlet node as the power node of the secondary network.

Step 2: Use the method in the previous section to calculate the node port voltage, node-to-ground voltage, branch current, line power, and line loss data of the secondary network.

Step 3: Substitute the power flow calculation results of the secondary network into (11), calculate the equivalent CPL value of the BDCT, and update the CRL admittance matrix and the CCL current column vector of the primary network node. Calculate the network node port voltage again using (18).

Step 4: Use (19), (21), and (24) to calculate the node port voltage, node-to-ground voltage, branch current, line power, and line loss data of the primary network.

In summary, the process of linearized power flow calculation of BDCDN with multiple flexible equipment can be summarized and described in Fig. 8.

5. Simulation results

5.1. Modified IEEE14-node

The modified IEEE14 node BDCDN with various types of flexible equipment is used for power flow calculation, voltage unbalance analysis, and error analysis to verify the correctness and validity of the linear power flow calculation of the BDCDN with various types of flexible equipment proposed in this study and the suppression effect of flexible equipment on the voltage unbalance phenomenon. The system structure is shown in Fig. 9.

The bipolar distribution network has two voltage levels: nodes 0–4 are ± 750 (1500) V, whereas nodes 5–13 are ± 400 (800) V. Node 0 is connected to the large power grid and is ± 750 (1500) V power node. Nodes 1–4 mainly bear the industrial and subway traction loads, and nodes 5–13 mainly bear the loads of electric vehicle charging stations, data centers, and residential electricity. The DCPFC is added to the positive and negative poles between nodes 2 and 3 (the positive and negative equivalent voltages V_{pC}, V_{nC} are both 5 V). The DCES is connected to the both positive and negative loads on node 10 (the output

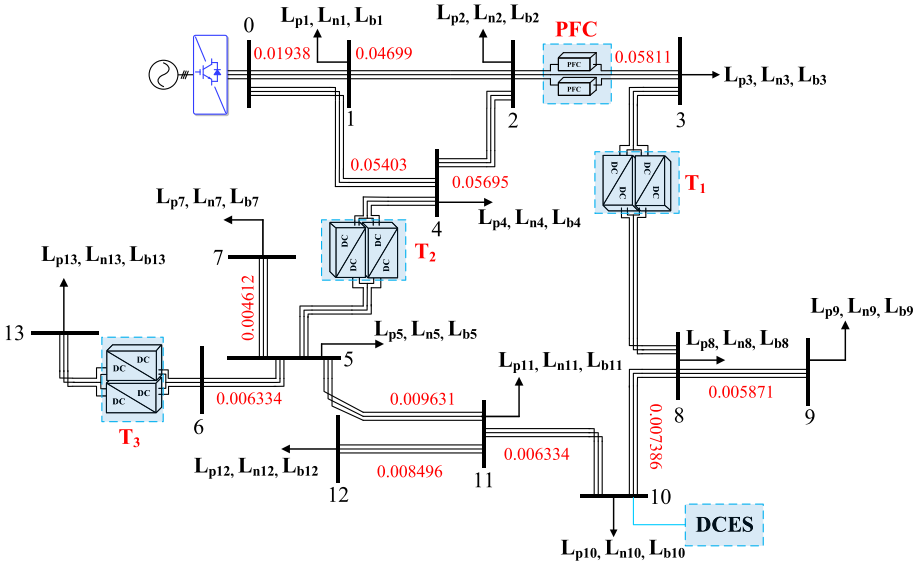


Fig. 9. Structure diagram of modified IEEE14-node bipolar DC microgrid example.

Table 1
Load parameter table of modified IEEE14-node bipolar DC microgrid.

Node	L_p			L_n			L_b		
	P_P/kW	Z_R/Ω	I_I/A	P_P/kW	Z_R/Ω	I_I/A	P_P/kW	Z_R/Ω	I_I/A
0	—	—	—	—	—	—	—	—	—
1	20.7	0	0	0	50	0	0	0	0
2	0	0	0	0	0	0	34.6	0	0
3	21.3	80	2	20.6	120	3.1	0	0	0
4	16.6	0	0	9.7	0	0	64.9	90	5.6
5	9.2	0	0	2	0	0	0	0	0
6	8.6	70	0	4.6	0	7.3	0	0	0
7	0	0	8.4	0	0	6.7	0	0	0
8	11.6	0	0	4.7	0	0	0	0	0
9	13.6	0	0	2.7	0	0	0	0	0
10	0	80	7.4	0	70	5.8	0	0	0
11	11.3	0	0	2.1	100	0	0	0	0
12	12	90	2	1.2	105	0	0	0	0
13	4.6	0	1	3.7	0	2	0	0	0

Table 2
Parameters of bipolar DCT.

BDCT	n_c	f_s/kHz	L/mH	Y_c/Ω	Control method	Control variable
T_1	1.875	10	0.4000	0.2	CPC	$d_{T1\ up}=0.3$ $d_{T1\ down}=0.2$
T_2	1.875	10	0.0015	0.2	CRC	$n_{T2\ up}=1.875$ $n_{T2\ down}=1.875$
T_3	1.000	10	0.0800	0.2	CVC	$V_{13pnu}=400$ V $V_{13nun}=400$ V

voltages V_{esp} and V_{esn} are both 200 V). The resistance parameters(Ω) of the positive, neutral, and negative lines are shown in the red numbers in Fig. 9. The ZIP load data of the positive, negative, and bipolar ports of each node are shown in Table 1. The transformer units in BDCTs T1–T3 adopt the form of DAB, and their parameters are shown in Table 2.

Matlab and Simulink are used to compile the linearized power flow calculation program and model simulation. Then, the power flow calculation and simulation results of the network are obtained. Table 3 shows the comparison of the simulation results and calculation results at each node port voltage. The maximum voltage calculation error is 0.073%, which is within the acceptable range. Therefore, the algorithm principle proposed in this study is correct and can be applied to a BDCDN with various flexible equipment.Table 4..

Table 3
Comparison of the simulation of the voltage calculation of each node port of IEEE 14 node case.

Node	Calculation/V		Simulation/V		Error/%	
	V_{pnu}	V_{nun}	V_{pnu}	V_{nun}	V_{pnu}	V_{nun}
0	750.000	750.000	750.000	750.000	0.000%	0.000%
1	746.679	748.916	746.679	748.915	0.000%	0.000%
2	740.529	746.392	740.516	746.390	0.002%	0.000%
3	741.704	748.792	741.663	748.792	0.005%	0.000%
4	738.150	747.206	738.142	747.203	0.001%	0.001%
5	383.199	394.969	383.476	394.721	0.072%	0.063%
6	382.872	394.844	383.150	394.593	0.073%	0.063%
7	383.153	394.946	383.429	394.698	0.072%	0.063%
8	380.223	396.234	380.494	395.987	0.071%	0.062%
9	379.843	396.363	380.114	396.117	0.071%	0.062%
10	380.730	395.876	381.002	395.629	0.072%	0.062%
11	381.257	395.650	381.530	395.403	0.072%	0.062%
12	380.675	395.855	380.947	395.608	0.072%	0.062%
13	400.000	400.000	400.000	400.000	0.000%	0.000%

Table 4
Table of computational time of 14-bus system.

Algorithm	φ = 1	φ = 0.8	φ = 1.2
Linearization	0.0198 s	0.0195 s	0.0201 s
Iteration	0.0763 s	0.0492 s	0.1097 s

The verify the effectiveness of the proposed model, the linearized algorithm is compared with the iterative algorithm proposed in [28] using the same 14-node system. The original Jacobian Matrix in [30] was modified according to the static models of flexible devices discussed in section 3, and below are the results of comparison. As shown in Fig. 10 and Fig. 11, the error levels of two algorithms are approximately the same, which can prove the effectiveness of proposed linearized algorithm.

In order to compare the computational speed between the existing iterative method and the proposed linearized method, the load magnitude percentage φ is introduced based on the 14-bus system. With different three φ values, three times of calculation are carried out by MATLAB, and the comparison results of computational time are shown in Table 1. From Table 1, we can see that the computational time of proposed linearized algorithm is 0.0198 s, 0.0195 s, 0.0201 s separately

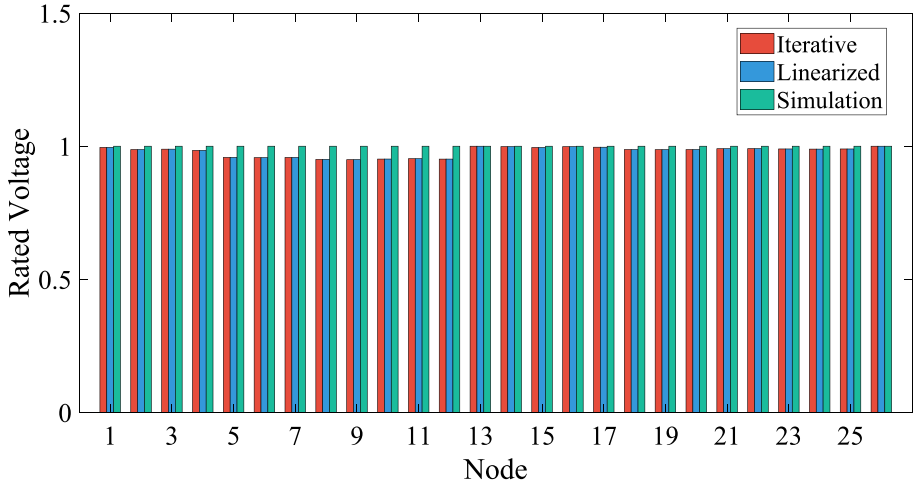


Fig. 10. Rated value of 2 algorithms and Simulink.

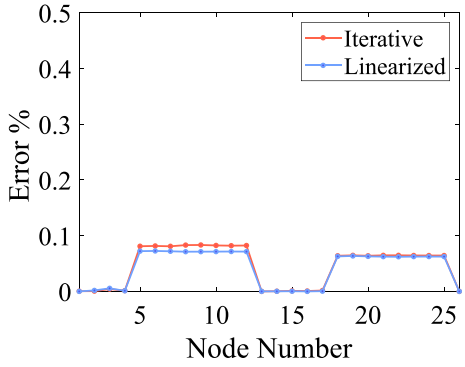


Fig. 11. Comparison of error values.

Table 5
Shift phase combination of positive and negative units of DCT under CPC.

BDCT	Group	Shift ratio group
T_2	1	$d_{up}=0.3$, $d_{down}=0.2$
	2	$d_{up}=0.3$, $d_{down}=0.1$

when φ is equal to 1, 0.8 and 1.2, which is basically unchanged. The computational time of iterative algorithm is 0.0763 s, 0.0492 s, 0.1097 s separately when φ is equal to 1, 0.8 and 1.2. It can be observed that the computational time increases with the increase of load magnitude. Under different load magnitudes, the voltage distribution of system is different, and the convergence direction and speed of the iterative algorithm are affected, so the computational time is different in three cases. On the contrary, the proposed linearized algorithm is not affected by that, so the computational time is basically unchanged, which are all much shorter than the iterative algorithm.

To verify the influence of the connection of flexible equipment on the voltage unbalance of the BDCDN, different flexible equipment connection schemes are set up, and the network voltage unbalance, voltage deviation, power loss and other indicators under different schemes are calculated to study the suppression effect of flexible equipment on voltage unbalance and voltage deviation. In this study, we use the voltage deviation and voltage unbalance indexes defined in the References [29–31].

To analyze the effect of BDCT on the improvement of voltage quality, the shift phase values of the positive and negative electrode units of the BDCT with CPC are changed, then obtain two combinations of different shift phases, as shown in Table 5. Using the power flow calculation method proposed in Section4.4, the calculation results of the voltage unbalance, voltage deviation and power loss of each node under the corresponding combination are obtained, as shown in Fig. 12.

According to Fig. 12(a)~(b), it can be observed that the change of the shift phase value of positive and negative electrode units of the BDCT with CPC can affect the voltage unbalance and voltage deviation of each node. Compared with group (1), group (2) reduces the shift phase value of the negative unit. In this case, the positive voltage deviation of each node basically did not change, but the absolute value of the negative voltage deviation of the secondary network composed of 5 ~ 12 nodes increased. However, the voltage unbalance of almost all nodes had a certain degree of decline. According to Fig. 12(c), it can be seen that the power loss in the group (2) was reduced to a certain extent.

To analyze the effect of DCPFC and DCES on voltage quality improvement, three flexible equipment connection cases are designed while keeping the original BDCT parameters unchanged, as shown in Table 6. Under the three connection cases, the power flow and voltage unbalance, voltage deviation and power loss were calculated, and the results are shown in Fig. 13 and Fig. 14.

According to the comparative analysis in Fig. 13 and Fig. 14, after adding flexible equipment, the voltage unbalance and voltage deviation of each node in the entire network are improved to a certain extent. In case 1, the connection of the DCPFC makes the positive and negative voltages of node 3 closer to the rated value, while suppressing the voltage unbalance and voltage deviation of node 3, and reducing the sum of voltage unbalance of each node from 32.33% to 31.60%, and the total power loss is also reduced. However, the effect of suppressing voltage unbalance and voltage deviation of other nodes is not obvious. In case 2, the connection of DCES reduces the voltage unbalance and positive voltage deviation of nodes 5 to 12, and reduces the power loss, but the absolute value of the sum of the negative voltage deviation of each node increase from 10.99% to 11.49%. In case 3, the simultaneous connection of DCPFC and DCES reduces the voltage unbalance of almost all nodes, which is better that case 2 in suppressing voltage deviation and power loss.

Based on the above analysis, it can be seen that the parameters of flexible equipment such as BDCT, DCPFC and DCES will affect the voltage unbalance, voltage deviation and power loss, and different flexible equipment have different control ranges and control sensitivities of different voltage indicators. It may also cause deterioration of the remaining indicators.

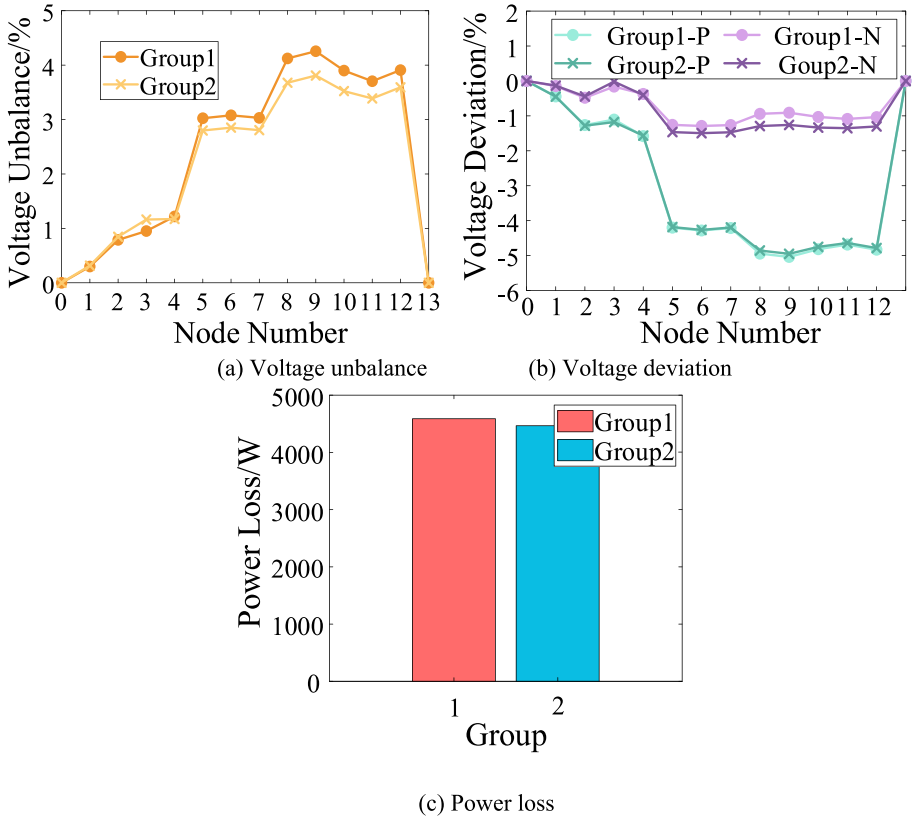


Fig. 12. Comparison of voltage unbalance, voltage deviation percentage of each node and total power loss under different combinations of shifts ratio.

Table 6
Flexible device access schemes.

Case	Equipment	Position and parameter
1	Install only PFC	Between node 2 and node 3 , $V_{pC} = 15$ V , $V_{nC} = 5$ V
2	Install only DCES	At node 10 , $V_{esp} = 200$ V, $V_{esn} = -200$ V
3	Install PFC and DCES	Same as above

5.2. *Modified IEEE33-node*

The modified IEEE33 node is used to verify the validity of the linear power flow model of the BDCDN with various type of flexible equipment proposed in this paper under more nodes, and the system structure is shown in Fig. 15. The system consists of 33 nodes and 34 branches. The network and load parameters are shown in Table 7. The network has two voltage levels: nodes 0–4 and 5–13 are ± 750(1500 V), whereas nodes 5–17 and 18–23 are ± 400(800 V). Node 0 is a balance node connected to the large AC power grid through the converter station with CVC, which means the port voltages of positive and negative poles of node 0 are both 750 V. Nodes 9 and 15 are connected to distributed power sources, and BDCTs are connected between nodes 4–5, 23–24 and 11–21 respectively to perform voltage level conversion or stable voltage control. The transformer units of BDCT1 ~ BDCT3 are all in the form of DAB. Node 30 is connected to DCES for voltage control of critical loads, and a series-type bipolar DCPFC is connected between nodes 17 and 32 to complete the power flow control of 400 V low voltage system. Parameters of BDCT, DCES and bipolar DCPFC are as shown in Table 8 to 10.

Matlab and Simulink are used to compile the linearized power flow calculation program and model simulation, and obtained the calculation and simulation results. According to calculation and simulation results of port voltages of each node, draw the results and errors as shown in Fig. 16. It can be observed that the node port voltage calculation errors of the entire network within the 10^{-3} level in the modified IEEE33 node model according to the Fig. 16. Because the network contains various types of flexible equipment, whose control methods are diverse, and the loss of internal power electronic devices is diverse, the steady-state model of flexible equipment proposed in this paper has a certain accuracy error. The maximum voltage calculation error occurs at the negative port of node 11, which is 0.186% and is in the acceptable range. Therefore, the principle of the algorithm propose in this paper is correct, and it can be applied to the BDCDN with more nodes and various types of flexible equipment.

To verify the influence of the connection of flexible equipment on the voltage unbalance of the BDCDN, different flexible equipment connection schemes are set up, and the network voltage unbalance, voltage deviation, and other indicators under different schemes are calculated to study the suppression effect of flexible equipment on voltage unbalance and voltage deviation.

To analyze the effect of BDCT, DCPFC and DCES on voltage quality improvement, change the shift phase value of the positive and negative units of BDCT with CPC and the types of flexible equipment connected, then five flexible equipment connection cases are designed, as shown in Table 11. Under the five connection cases, the power flow and voltage unbalance, voltage deviation were calculated, and the results are shown in Fig. 17.

According to Fig. 17, it indicates that changing the shift phase value of the positive and negative units of BDCT with CPL and changing the types of flexible equipment connected can affect the voltage unbalance and voltage deviation at each node. Compare with case 1, case 2 reduces the shift phase value of the negative unit. In this case, the absolute value of the positive voltage deviation of each node decrease while the absolute value of the negative voltage deviation increases, but the voltage unbalance of almost all nodes have a certain of decline. In case 3, the connection of DCPFC makes the absolute value of the positive and

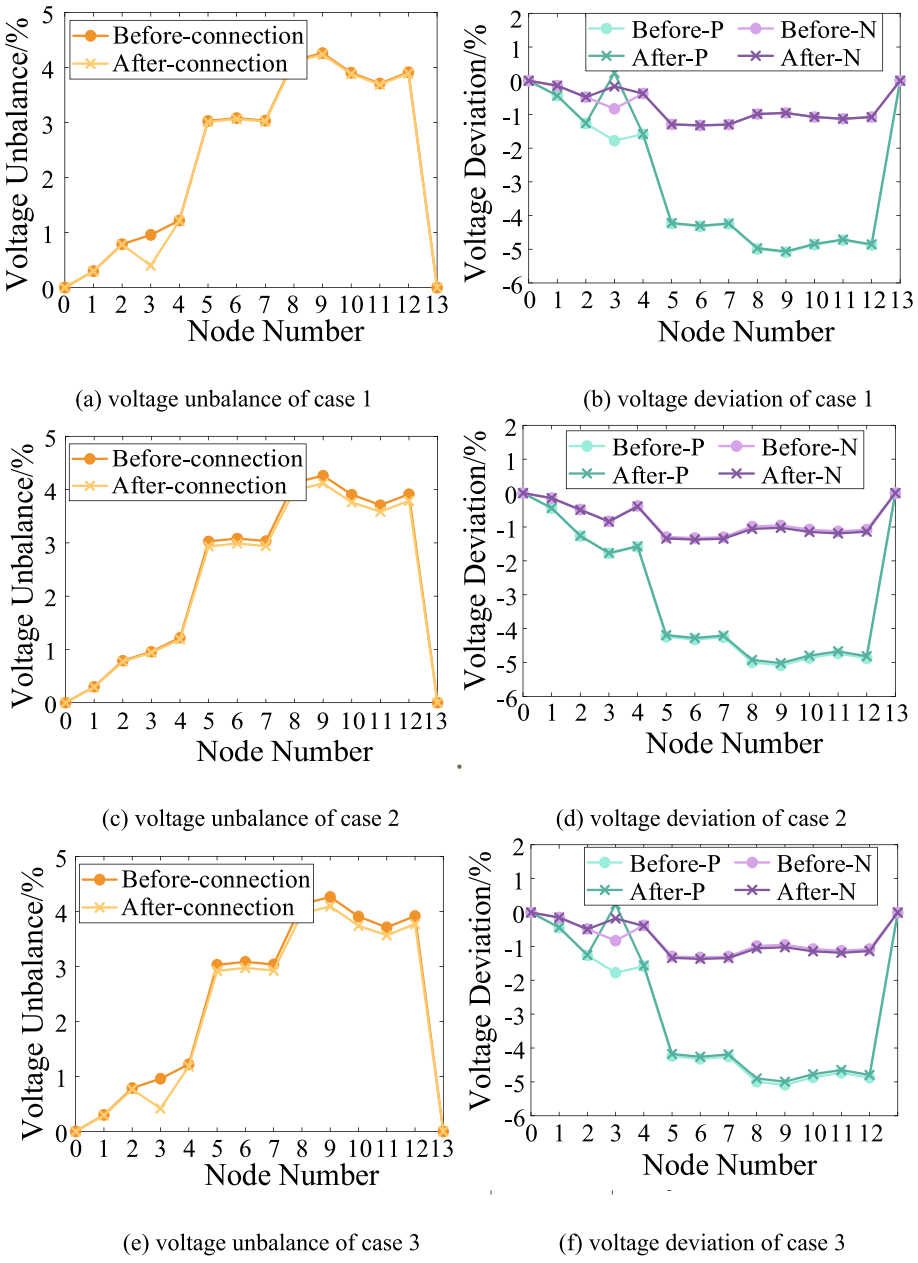


Fig. 13. Comparison of port voltage, voltage unbalance and voltage deviation percentage under different flexible equipment connection cases.

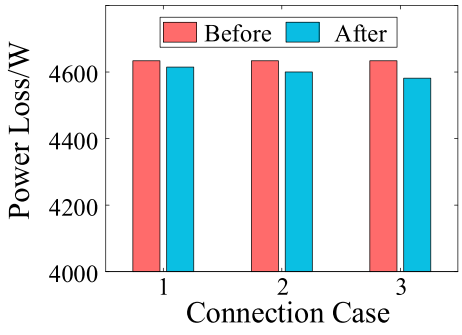


Fig. 14. Comparison of power loss under flexible equipment connection cases.

negative voltage deviation of node 17 and nearby nodes decrease, but the voltage unbalance increases to a certain extent, and the absolute value of the positive and negative voltage deviation of node 32 and nearby nodes increase, but the voltage unbalance decrease to a certain extent. In case 4, the connection of DCES reduces the absolute value of the positive voltage deviation and the voltage unbalance of all nodes, but increases the sum of the absolute value of negative voltage deviation of each node from 95.82% to 98.86%. In case 5, the simultaneous connection of DCPFC and DCES reduces the voltage unbalance of almost all nodes, which is better than case 4 in suppressing the voltage deviation of positive and negative electrodes.

To further explore the impact of the connection of flexible equipment on its connection nearby nodes, nodes 3 and 4 near the BDCT, and the nodes 15,16,17,30,31,32 near the DCES and DCPFC are selected for research. Observe the influence of different flexible equipment parameters and connection cases on the voltage deviation and voltage unbalance as shown in Fig. 18.

According to Fig. 18, in case 3, the connection of DCPFC between

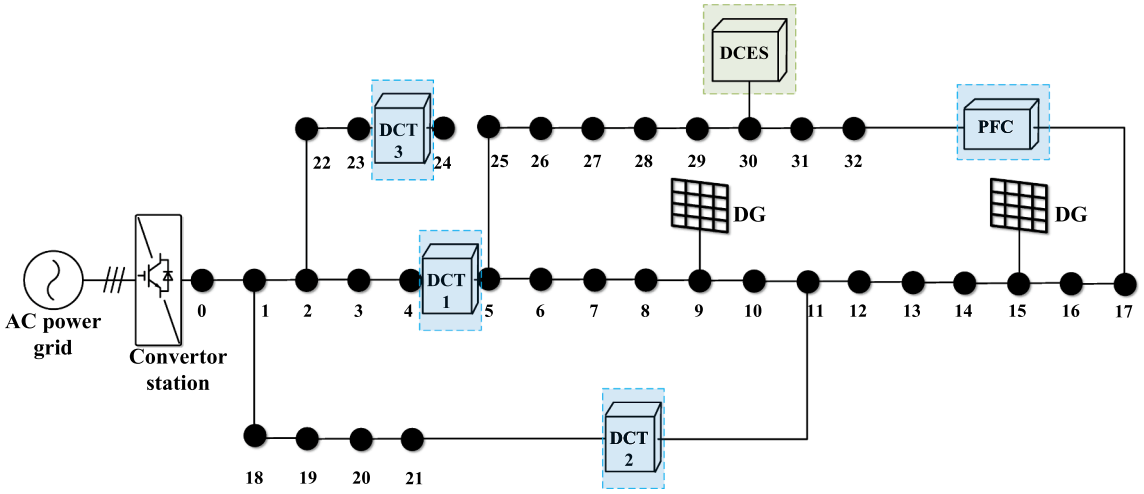


Fig. 15. Structure diagram of modified IEEE33-node bipolar DC microgrid.

Table 7
Load parameter table of modified IEEE33-node bipolar DC microgrid.

Branch	Start node	End node	resistance /Ω	Load power of end node/ kW	Polarity of load	Branch description
1	0	1	0.0461	10	positive	ordinary
2	1	2	0.0493	13	negative	ordinary
3	2	3	0.0366	12	positive	ordinary
4	3	4	0.03811	24	bipolar	ordinary
5	4	5	/	/	/	DCT
6	5	6	0.001872	6	positive	ordinary
7	6	7	0.007114	9	negative	ordinary
8	7	8	0.0103	6	negative	ordinary
9	8	9	0.01044	−5	negative	ordinary
10	9	10	0.001966	9	positive	ordinary
11	10	11	0.003744	6	negative	ordinary
12	11	12	0.01468	6	negative	ordinary
13	12	13	0.005416	8	positive	ordinary
14	13	14	0.00591	6	negative	ordinary
15	14	15	0.007463	−11	positive	ordinary
16	15	16	0.01289	6	positive	ordinary
17	16	17	0.00732	9	positive	ordinary
18	1	18	0.082	5	positive	ordinary
19	18	19	0.07521	11	negative	ordinary
20	19	20	0.020475	15	negative	ordinary
21	20	21	0.035445	12	positive	ordinary
22	2	22	0.02256	18	bipolar	ordinary
23	22	23	0.0449	6	positive	ordinary
24	23	24	/	9	positive	DCT
25	5	25	0.00203	7	positive	ordinary
26	25	26	0.002842	3	negative	ordinary
27	26	27	0.01059	6	positive	ordinary
28	27	28	0.008042	9	negative	ordinary
29	28	29	0.005075	5	negative	ordinary
30	29	30	0.009744	8	positive	ordinary
31	30	31	0.003105	6	negative	ordinary
32	31	32	0.00341	9	positive	ordinary
33	17	32	0.00264	/	/	PFC
34	11	21	/	/	/	DCT

node 17 and 32 makes the absolute value of the positive and negative voltage deviation of nodes near node 17 decrease, but the voltage unbalance increase, and makes the absolute value of the positive and negative voltage deviation of nodes near node 32 increase, but the voltage unbalance decrease. In case 4, the connection of DCES at node 30 reduces the voltage unbalance and absolute value of the positive voltage deviation, but increase the absolute value of the negative voltage deviation. In case 5, the simultaneous connection of DCES and DCPFC reduces the voltage unbalance of almost all nodes, and has a better suppression effect on the positive and negative voltage deviation of nodes near node 17.

Compared with original case 1, case 5 reduces the voltage unbalance of almost all nodes, but it still have a higher voltage unbalance than case 2 and case 3. That's because the connection of DCES will decrease the voltage unbalance while the connection of DCPFC will increase the voltage unbalance, and they will have a coupling influence when they are both connected.

Based on the above analysis, it can be seen that the parameters of flexible equipment such as BDCT, DCPFC and DCES will affect the voltage unbalance, voltage deviation and power loss, and different flexible devices have different control ranges and control sensitivities for different voltage indexes. Therefore, when all flexible devices are connected to the power grid at the same time, the voltage coupling control

Table 8
Parameters of DCT1 ~ DCT3.

BDCT	n_c	f_s/ kHz	L/mH	Y_c/ Ω	Control method	Control variable
DCT1	1.875	10	0.4000	0.2	CRC	$n_{DCT1\ up} = 1.875$ $n_{DCT1\ down} = 1.875$
DCT2	1.875	10	0.0015	0.2	CPC	$d_{DCT2\ up} = 0.35$ $d_{DCT2\ down} = 0.25$
DCT3	1.000	10	0.0800	0.2	CVC	$V_{24pnu} = 400$ V $V_{24nun} = 400$ V

Table 9
Parameters of bipolar DCPFC.

Start node of connection	End node of connection	Positive controlled voltage V_{pC}/V	Negative controlled voltage V_{nC}/V
32	17	5	5

Table 10
Parameters of DCES.

Node	Positive DCES		Negative DCES	
	Output voltage V_{esp}/V	NCL resistance /Ω	Output voltage V_{esn}/V	NCL resistance /Ω
30	200	100	−200	100

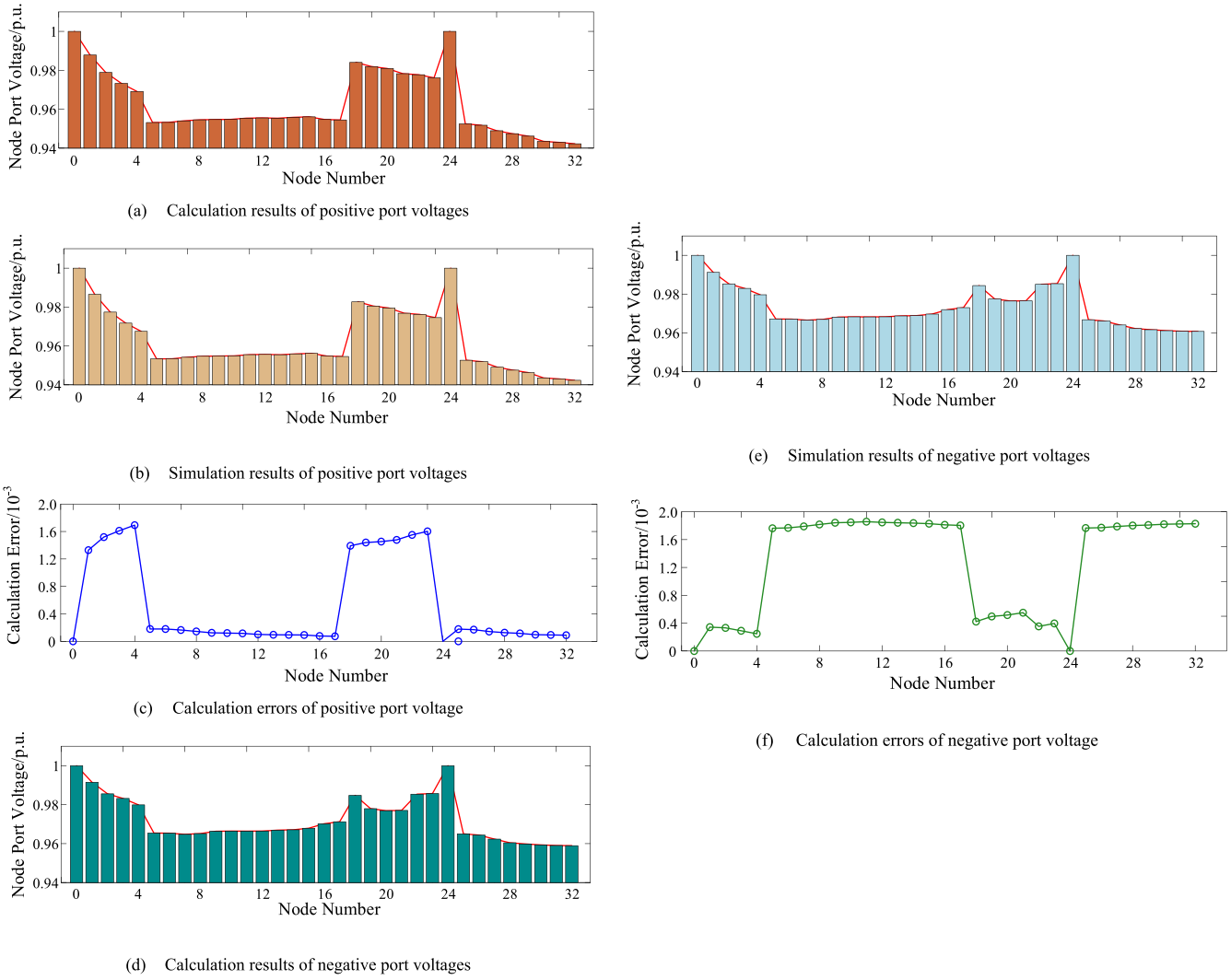


Fig. 16. Calculation results of IEEE 33 node case.

Table 11
Flexible device access schemes.

Case	Equipment	Position and parameter
1	Install only DCT	$d_{up} = 0.35$, $d_{down} = 0.25$
2	Install only DCT	$d_{up} = 0.35$, $d_{down} = 0.15$
3	Install DCES and DCT	At node 10, $V_{esp} = 200$ V, V_{esn} = -200 V
4	Install PFC and DCT	Between node 2 and node 3, V_{pC} = -3V, V_{nC} = -5V
5	Install DCPFC,DCES, DCT	Same as above

effect will be generated, and the deterioration of other indicators may be caused when the voltage unbalance is decreased.

5.3. Error analysis

During the process of building models, there exists some errors which come from the modeling of flexible equipment and the modeling of CPL linearization. The error resource contains following four aspects:

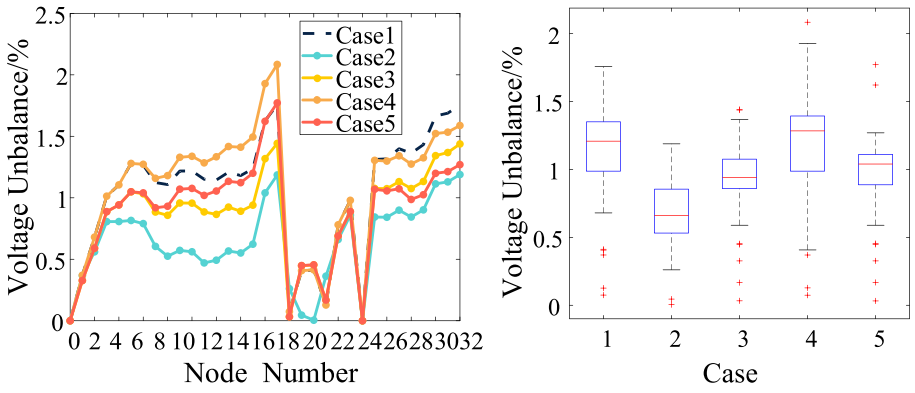
(1) Error of linearization of CPL model.

The transformation of CPL to a parallel structure with a CRL and a CCL is indeed an error source. According to the CPL linearization model proposed in this paper, the relative error value in the power equation linearization process can be deduced. For the positive CPL of node 1, its relative voltage error value is shown in (25).

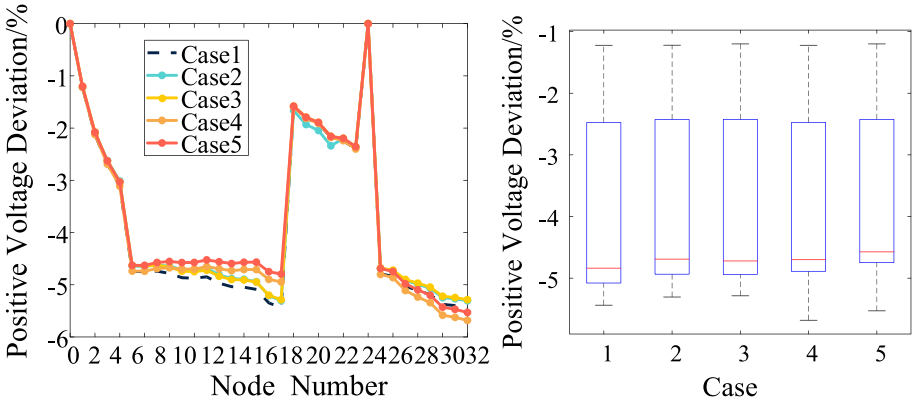
$$\left|\frac{V_{pnu1} - V'_{pnu1}}{V_{pnu1}}\right| \times 100\% = = \frac{V_{pnu1} - \frac{1}{\left|\frac{1}{V^0_{pnu1}} - \left(\frac{1}{V^0_{pnu1}}\right)^2 \left(V_{pnu1} - V^0_{pnu1}\right)\right|}}{V_{pnu1}} \times 100\% = \left|\frac{-V^2_{pnu1} + 2V^0_{pnu1}V_{pnu1} - \left(V^0_{pnu1}\right)^2}{-V^2_{pnu1} + 2V^0_{pnu1}V_{pnu1}}\right| \times 100\% \quad (25)$$

According to (25), the function relationship between the actual operating voltage and the error of the CPL linearized model can be drawn, as shown in Fig. 18. It can be seen from Fig. 18 that the error of the CPL linearization model increases with the increase of the actual operating voltage deviation. When the maximum deviation value is 5%, that is, the actual operating voltage unit value is 0.95/1.05, the error value of the CPL linearization model is 0.2506%. Considering the feature of no iterative calculation, the error value of the CPL linearization model proposed in this paper is small and within the acceptable range.

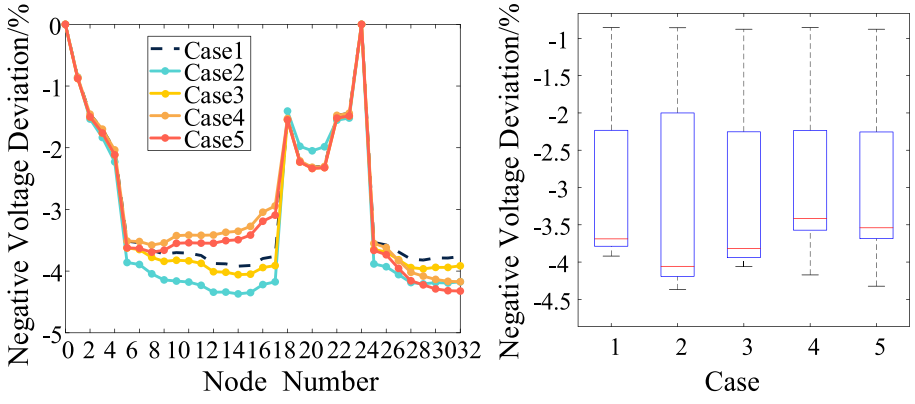
To prove that, we built two simulation girds, which contain a gird without DCT and the other gird with DCT. In both grids, we draw the tendency chart of maximum voltage solution, which changes with the



(a) Voltage unbalance



(b) Positive voltage deviation



(c) Negative voltage deviation

Fig. 17. Comparison of voltage unbalance and voltage deviation percentage under different flexible device access schemes.

increase of the number of load nodes containing only CPL. And the error shown is obtained by comparing the results calculated by the algorithm proposed in this paper with the simulation results of Simulink, as shown in Fig. 19a.Fig. 19b..

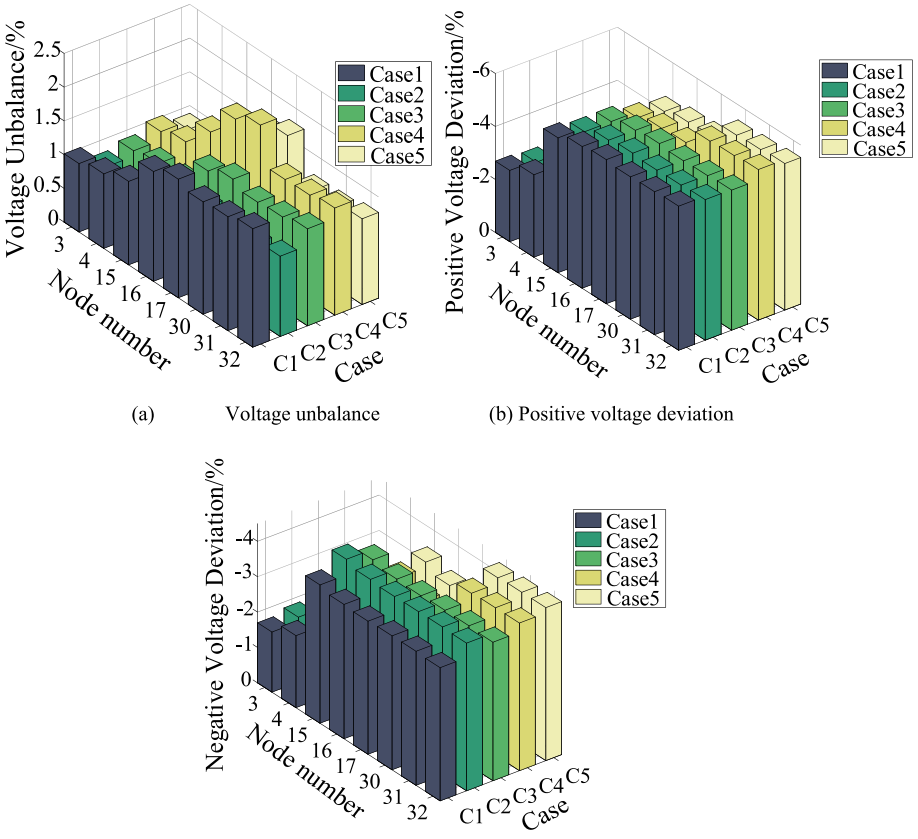
We can observe that the error will increase exponentially when there is no DCT. After the addition of DCT, the change will not show an obvious exponential tendency due to the control function of DCT and the accuracy of DCT model. Therefore, the phenomenon that the error of IEEE33 is twice as much as the error of IEEE14 is within the normal range.

(2) Error of equivalent modeling of DCT port transfer characteristics.

In this paper, the equivalent loss resistance is used to simulate the steady-state operation loss of DCT when establishing the equivalent high-low voltage transfer model of DCT. The loss caused by dynamic change of inductor and current and the loss of high-frequency switching loss are not taken into account, so there are certain errors in the calculation of power flow.

Reference [32] established a DAB steady-state port transmission model which considered the dynamic change of inductor and current and high-frequency loss, and the accuracy was verified by simulation.

To analyze the error of the equivalent model of dynamic effect characteristics of DCT port transmission established in this paper, the DAB steady-state port transmission model established in reference [32]



(a) Voltage unbalance (b) Positive voltage deviation

(c) Negative voltage deviation

Fig. 18. Comparison of voltage unbalance and voltage deviation percentage of connection nearby under different flexible device access schemes..

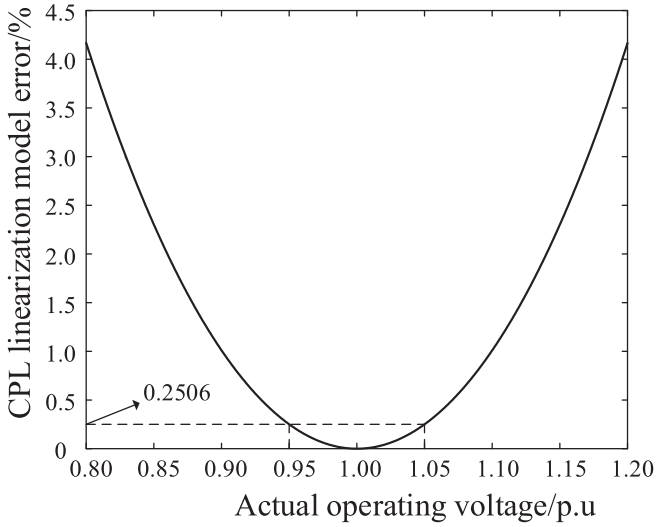
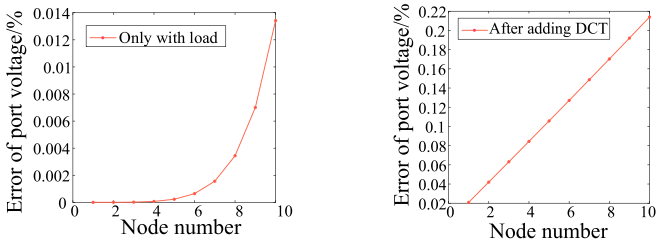


Fig. 19a. a. Actual operation error diagram of CPL linearized model.



(a) Grid with only connecting CPLs (b) Grid with adding DCT

Fig. 19b. Error of simulated voltage value and calculated voltage value.

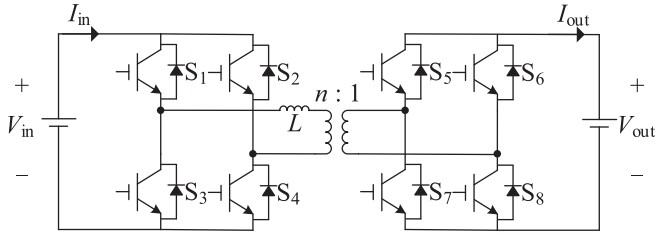


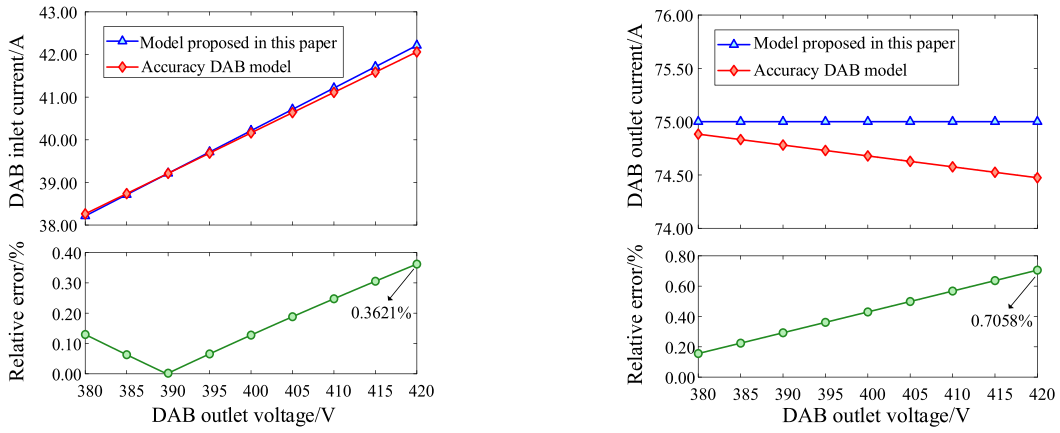
Fig. 20. The topology of DAB.

was selected as the benchmark to conduct error simulation analysis for the DCT model.

For the DAB model shown in Fig. 20, under the control of single phase shift, the exact model of voltage and current port characteristics on both sides of DAB considering the loss can be built. And compare our model in this paper with the accuracy DAB model, the DAB inlet and outlet current and their relative error values under two models were obtained, as shown in Fig. 21.

(3) Error of DCPFC.

Based on the two topologies of voltage type DCPFC, we deduced the transfer characteristics of its high and low side ports in this paper. Under the topology structure of DCPFC of DC-DC converter type, we



(a) DAB inlet current calculation value and relative error (b) DAB outlet current calculation value and relative error

Fig. 21. Calculated dab inlet current values and their relative errors under the two models.

equivalents it to DCT. Under this equivalent, the steady-state modeling error is mainly manifested as the error caused by the loss of the dynamic change of inductor current and loss of high-frequency switching that is not considered in DCT. This partial error analysis is the same as the error analysis of DCT above. Under the topology of DCPFC of adjustable voltage type, an equivalent model based on parallel current source is derived according to its port characteristics, and the power flow model is used to modifying and access influence. According to the internal topology structure of DCPFC, there existed control error because the DCPFC of adjustable voltage type is composed of capacitor and transistors, and the output port voltage of the adjustable voltage type DCPFC can not be completely the same as the control target value during its steady operation, so the DCPFC equivalent model based on parallel current source has accuracy error in power flow calculation. However, the error varies with the control error of the adjustable voltage type DCPFC, and it is 0 when the control error is 0. Therefore, the error of the equivalent model of DCPFC port transfer characteristics is generally small and within the acceptable range within the normal operating range of voltage.

(4) Error of DCES.

In this paper, according to the operation principle of DCES, the characteristics of its port are derived, and it is equivalent to ZIP load with certain value of impedance, current and power. In the process of building the electric spring model, the errors are mainly manifested as the change of capacitance and inductance in the internal topology of the DCES and the linearization error of the CPL model.

The internal topology of the DC electric spring contains electric equipment such as capacitance and inductance. For the transient period, the output voltage Ves of the DC power spring is controlled by the change of the charge and discharge of the capacitor and inductance. Therefore, in the steady state period, the output voltage Ves is a fixed value, and the error comes from the internal of the electirc spring. This paper focuses on the fixed voltage output of the electric spring, so this error will not affect the calculation of the voltage value in the establishment of the model. And the CPL model is approximately treated when the CPL is linearized, so when the DCES modeling is equivalent to the CPL, there exists errors in the linearized process.

6. Conclusion

This study analyzes and studies the control methods and power flow calculation of a BDCDN with multiple types of flexible equipment. The key conclusions are summarized as follows:

1) The BDCDN model is established by decomposing the network for nodal analysis. The CPL is linearized at the working point by Taylor expansion. Then, the load linearization model is established to reduce the iterations of power flow calculation.
2) The influence of the BDCT, DCPFC, and DCES on the power flow calculation during the steady-state operation is analyzed. The voltage and current transfer models of these three types of flexible equipment are established and demonstrated to be effective.
3) A linearized power flow calculation method for a BDCDN with multiple types of flexible equipment is proposed. The validity and correctness of the proposed power flow algorithm are verified through the 14-node and 33-node BDCDNs with multiple types of flexible equipment, by comprising with existing iterative algorithm. The suppression of voltage unbalance through multiple types of flexible equipment is also verified.

In summary, the power flow calculation method proposed in this study provides an idea for solving the rapid noniterative power flow calculation of the DC distribution network with multiple types of flexible equipment under the bipolar structure. It also lays a foundation for subsequent DC distribution network planning, construction, and operation control research.

CRediT authorship contribution statement

Hongbin Wang: Data curation, Methodology, Formal analysis, Software, Validation, Writing – original draft, Visualization. **Niancheng Zhou:** Supervision, Project administration, Conceptualization. **Yu Zhang:** Conceptualization, Supervision, Methodology, Software, Investigation. **Jianquan Liao:** Writing – review & editing, Conceptualization, Investigation, Funding acquisition. **Shuyin Tan:** Data curation, Software, Formal analysis. **Xuan Liu:** Data curation, Software, Formal analysis. **Chunsheng Guo:** Software, Supervision, Formal analysis. **Qianggang Wang:** Supervision, Methodology, Validation, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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